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Comparative analysis of deep learning and traditional methods for IoT botnet detection using a multi-model framework across diverse datasets

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The proliferation of Internet of Things (IoT) devices has created unprecedented cybersecurity vulnerabilities, with botnets emerging as a critical threat to network infrastructure. This study focuses on traditional machine learning and deep learning approaches, proposes a novel ensemble framework to address these issues, integrating Convolutional Neural Network (CNN), Bidirectional Long Short-Term Memory (BiLSTM), Random Forest (RF), and Logistic Regression (LR) via a weighted soft-voting mechanism. Our approach introduces a Quantile Uniform transformation to reduce feature skewness, a multi-layered feature selection method to enhance discriminative power, an individual performance of deep learning–traditional machine learning and a hybrid models (ensemble models) for robust detection. Evaluated on BOT-IOT, CICIOT2023, and IOT23 datasets, the framework achieves 100% accuracy on BOT-IOT, 99.2% on CICIOT2023, and 91.5% on IOT23, outperforming state-of-the-art models by up to 6.2%. These contributions advance IoT security by enabling scalable, high-performance detection adaptable to diverse network scenarios, with practical optimizations for real-world deployment.

Keywords IoT, Cyber-attacks, Botnet, Deep learning, Traditional machine learning, SMOTE

The Internet of Things (IoT) represents a transformative technological paradigm that has fundamentally altered our digital landscape[1]. While this interconnectivity enables smart cities, healthcare systems, and industrial applications[2], it simultaneously creates unprecedented cybersecurity vulnerabilities. Botnets pose a major threat to IoT security as traditional detection methods fail to evolve at the same rate as the advancing technologies and tactics used by hackers[3]. Although the current detection approaches provide satisfactory accuracy, they are lacking in two important aspects: the inability of the existing detection approaches to be thoroughly validated using multiple datasets as well as the failure to consider other important performance measures that are not solely based on accuracy[4].

Currently, the field of botnet detection continues to evolve due to the application of advanced techniques, including deep learning and ensemble methods for detection. Although these state-of-the-art methods provide high accuracy by using deep learning neural networks and sophisticated advanced feature engineering[5]. The implementation of these methods in real-time detection systems has been found to be very effective for identifying complex attack patterns. However, there is still a major challenge: while the current automated response mechanisms produce favorable outcomes in protecting the IoT networks[6], the problem is that these

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solutions are usually characterized by complex algorithms which rely on inadequate data and are not followed by proper data cleaning and validation procedures[7].

The Current botnet detection research faces several key issues notwithstanding the recent progress. Although the current state-of-the-art methods using deep neural networks and ensemble learning provide high accuracy metrics, as pointed out in[8], they tend to ignore other important performance measures like MSE, RMSE, and MAE, which are more relevant for assessing the performance in the real world[8]. Furthermore, training models on the individual dataset's limits evaluating the generalizability of these models across diverse IoT environments, thus casting a doubt on their real-world usability[9]. Third, most of the preprocessing techniques do not solve the problems of skewness, class imbalance, and feature selection when used all together[10].

The performance of the classifiers also depends on the type of dataset used to train the model. BOT-IOT[11], CICIOT2023[12] and IOT23[13] are some of the benchmark datasets that are used to compare the performance of various models for IOT IDS and IOT attack detection. These datasets offer different approaches to studying IoT network attacks. BOT-IOT is a large, simulated dataset focused on network forensics, while CICIoT2023 is a real-time dataset created using an extensive IoT topology, and IOT-23 captures real-world IoT traffic from specific devices.

Our research presents a structured way of improving the method for botnet detection as implemented in our paper, which involves five key methodological stages. First, we consider the problem of data quality in three different datasets, namely BOT-IOT, CICIOT2023, and IOT23, focusing on missing values, duplications, and feature skewness. Second, we implement a comprehensive skewness reduction framework, comparing multiple transformation methods (log, square root, yeo-johnson, quantile) with Quantile Uniform transformation emerging as optimal for preserving data integrity[14]. Third, we employ advanced feature selection through correlation analysis, Chi-square statistics with p-value validation, and feature dependency examination across label classes using advanced distribution and proportional analysis techniques. Fourth, we establish a robust model fitting optimization framework using Random Forest and Logistic Regression models with threshold-based decision-making to balance underfitting and overfitting. Fifth, we implement cross-validation techniques revealing dataset-specific challenges, addressing class imbalance through SMOTE[10], and comparing balanced versus PCA-transformed data approaches.

Our methodology demonstrates significant performance variations across datasets. The framework employs multiple validation techniques across different network environments, with cross-validation results revealing distinct patterns: BOT-IOT and CICIOT2023 demonstrate excellent classification characteristics with consistent performance across folds, while IOT23 presents more complex challenges that mirror real-world scenarios. Deep learning models (CNN with optimized layers, BiLSTM with tuned memory units) demonstrate superior accuracy but require substantial computational resources, while traditional approaches (Random Forest with optimized tree depth, Logistic Regression with regularization) offer remarkable efficiency with significantly lower computational overhead[15]. ML models producing top results in classification and anomaly detection tasks and have been widely applied to various real-world problems[16,17]. The systematic comparison between SMOTE balancing and PCA-based dimensionality reduction consistently validates the superiority of our balanced dataset approach across all evaluation metrics. The ensemble model provides insight into the decision-making process[18], and hyperparameters for the CNN, BiLSTM, Random Forest (RF), and Logistic Regression (LR) models are carefully tuned and integrated into the ensemble framework to optimize performance.

The study provides comprehensive experimental validation through:

- Systematic preprocessing and feature selection optimization
- Cross-validation based model assessment across multiple datasets
- Comparative analysis of traditional and deep learning approaches
- Extensive evaluation using classification, regression, and timing metrics
- Assessment of computational efficiency in various deployment scenarios
- Statistical validation across diverse attack patterns

Research contributions

The article presents the following main contributions:

1. Multi-Dataset Framework integrating BOT-IOT, CICIOT2023, and IOT23 for robust, generalizable botnet detection across diverse environments
2. Quantile Uniform Transformation application reducing feature skewness while preserving attack signatures (achieving near-zero skewness: 0.0003 vs. 1.8642 for log transformation)
3. Multi-Layered Feature Selection combining correlation analysis, Chi-square statistics with p-value validation, and distribution analysis to enhance discriminative power
4. Model Fitting and Optimization Framework with threshold-based decisions and cross-validation, achieving exceptional performance (100% accuracy on BOT-IOT, 99% on CICIOT2023, 85-90% on IOT23)
5. Weighted Voting Ensemble combining deep learning and traditional models for superior performance on complex datasets (100% accuracy on BOT-IOT, 99.2% on CICIOT2023, 91.5% on IOT23)
6. Comprehensive Evaluation Framework integrating classification, regression, and computational efficiency metrics for nuanced performance assessment
7. Practical Deployment Insights demonstrating SMOTE-balanced datasets' superiority for preserving attack patterns in real-world implementations

The remainder of this paper is organized as follows: Sect. Related work examines the related work, Sect. Proposed methodology proposed Methodology, Sect. "Experiments and results" presents comprehensive experimental results, Sect. "Discussion" discussion across all datasets and Sect. "Conclusions" conclusions.

Related work

The Internet of Things (IoT) has revolutionized industries such as healthcare, smart cities, and industrial systems, enabling seamless connectivity and data exchange. However, the rapid expansion of IoT networks has introduced significant cybersecurity challenges, with botnets emerging as one of the most prominent threats. These botnets exploit IoT vulnerabilities to orchestrate distributed denial-of-service (DDoS) attacks, data breaches, and other malicious activities. Consequently, the development of intrusion detection systems (IDS) leveraging machine learning (ML) and deep learning (DL) methodologies has become a critical area of research. This review critically examines advancements in Data Preprocessing and Quality Enhancement, Feature Selection and Optimization, Model Fitting and Cross-Validation, Class Balancing and Performance Optimization, Model Evaluation and Metrics focusing on datasets like BOT-IOT, CICIOT2023, and IOT23.

Data preprocessing and quality enhancement

IoT networks connect devices through sensors, gateways, and routers, generating massive volumes of data that require sophisticated preprocessing. While existing research has established basic preprocessing frameworks[19,20], critical gaps remain in handling the dynamic and complex nature of IoT security data. Current approaches often fail to address the challenges of real-time data volatility and the interconnected nature of IoT environments. A significant challenge in IoT security datasets lies in handling outliers and skewed distributions. Research demonstrates that traditional outlier detection methods[21,22] often fail to distinguish between legitimate anomalies (actual attacks) and statistical outliers, potentially leading to the removal of critical attack signatures. Studies show that while techniques like IQR and z-score are commonly used, they may not be appropriate for security datasets where extreme values could represent genuine attack patterns[23].

The literature reveals critical limitations in addressing skewness across multiple datasets. While various transformation techniques (Box–Cox, Yeo-Johnson) have been proposed[24,25], research shows that their effectiveness varies significantly across different IoT security datasets. Studies indicate that existing approaches often fail to:

1. Preserve important attack signatures during transformation
2. Maintain consistency across multiple datasets
3. Handle multi-modal and complex distributions effectively

The Bot-IoT dataset analysis[26,27] highlights significant challenges in standardization, particularly when dealing with skewed distributions and outliers. Current research demonstrates that standard preprocessing pipelines[28,29] often struggle to balance between normalizing features and preserving attack patterns. This becomes especially critical when working with multiple datasets like BOT-IOT, CICIOT2023, and IOT23.

Studies emphasize that existing standardization techniques[30,31] face significant limitations when applied to security datasets. The literature reveals that current approaches often:

- Fail to adapt to dataset-specific characteristics
- Struggle with maintaining consistency across multiple datasets
- Inadequately handle the relationship between outliers and attack patterns

Research Gaps and Future Directions:

1. Methodological Gaps:

- Limited investigation into adaptive outlier detection methods specifically designed for security datasets
- Insufficient research on transformation techniques that preserve attack signatures
- Lack of comprehensive frameworks for handling skewness across multiple datasets

2. Technical Limitations:

- Current standardization approaches[32–34] often compromise between outlier handling and information preservation
- Existing transformation methods may not adequately address the unique characteristics of IoT security data
- Limited understanding of the impact of different preprocessing techniques on model performance

3. Future Research Needs:

- Development of context-aware outlier detection methods
- Investigation of adaptive transformation techniques for security datasets
- Creation of unified preprocessing frameworks that can handle multiple datasets while preserving critical attack signatures
- Integration of domain knowledge in preprocessing decisions

This critical analysis reveals the need for more sophisticated approaches to handling outliers and skewness in IoT security datasets, particularly when working with multiple datasets simultaneously. Future research

should focus on developing methods that can maintain the delicate balance between data normalization and the preservation of security-relevant information.

Feature selection and optimization

Feature selection in IoT security datasets encompasses various types of features, including quantitative, qualitative, n-gram, network, and packet features[23]. The literature identifies significant challenges in handling these features, particularly regarding redundancy, noise, and irrelevant data that can degrade detection accuracy. Research demonstrates that effective feature selection strategies are crucial for addressing these performance-hindering factors[35]. Saheed et al[36] presented an IDS that used an autoencoder to reduce dimensionality of features after training on network flow data with Deep Convolutional Neural Network (DCNN) and LSTM. As LSTM performed better than DCNN in anomaly detection and provide a cost-effective, unsupervised solution.

A critical analysis of existing literature reveals several approaches to feature selection. Shetye[37] emphasize three fundamental methodologies: filter-based approaches for correlation evaluation, wrapper-based methods utilizing machine learning algorithms, and embedded-based techniques employing regularization. While these approaches provide a foundation, they often lack comprehensive statistical validation. Studies by Pilnenskiy[38] highlight how traditional feature selection methods, such as SelectKBest, may overlook important relationships between features and attack patterns.

Recent research has demonstrated promising results through various machine learning techniques. Studies show significant improvements in detection accuracy, with methods achieving up to 97.506% accuracy for DDoS attack detection[39]. Advanced approaches using evolutionary algorithms have shown even more promising results, achieving 99.98% accuracy and 99.63% F1 score[40]. Furthermore, innovations in feature selection through extra tree classifiers[41] and hybrid approaches combining autoencoders with optimization techniques[42] demonstrate the potential for more sophisticated feature analysis.

However, there are some important gaps in the current feature selection techniques:

1. The current approaches do not pay much attention to the interdependence of features across different attack types.
2. The current feature selection techniques do not adequately validate the statistical significance of the selected features.
3. There is also an insufficient analysis of the data across multiple data sets.
4. There is also insufficient consideration of the features' correlations in security contexts.

These limitations underscore the need to develop more sophisticated feature selection frameworks that incorporate statistical testing and validation processes while keeping in mind security relevant characteristics.

Model fitting and cross-validation

The current research in IoT security has shifted its focus on the role of appropriate model fitting and validation techniques. Author Zhu's[43] presents a comprehensive analysis of the consequences of overfitting optimization and how the traditional methods struggle to overcome this issue to find the best fit of the model, especially when dealing with complex security situations. The study also reveals that the conventional methods used for model fitting are not efficient in handling the problems of underfitting, which leads to missing threats, and overfitting, which results in the generation of false alarms.

There is a recent work by Prajapati and Singh where they propose an optimal solution approach for managing overfitting and underfitting in complicated systems, such as smart homes and other IoT devices and systems, by introducing a new framework[44]. The study highlights that even though these traditional metrics are crucial, they may not be enough to measure how well a model can handle various attacks. Gygia et al.'s[45] review on predictive overfitting shows that to achieve efficient model fitting, several factors has to be considered:

- Systematic validation across multiple datasets
- Clear threshold-based decision mechanisms
- Comprehensive performance analysis
- Balanced detection capabilities

Cohen et al.'s[46] in their research, 'Cross-Validation Conformal Risk Control' found out that there are huge differences in model performance when validated on different security datasets. Their study shows that the characteristics of the dataset used can greatly affect the performance of the model, especially where attack intelligence and patterns vary. This paper also stresses the need to use more robust validation techniques that can be flexible according to the security environment.

However, a critical analysis by Iyengar et al.[47] questions the notion of cross validation as the ultimate method for assessing evaluation of models and identifies several major issues with the current practices:

1. Inadequate focus on the problems of model fitting with respect to datasets.
2. Poor consideration of how model complexity interacts with the constraints of system security requirements.
3. Insufficient methods for assessing performance across several data sets.
4. Insufficient methods for the optimization of the threshold value.

Class balancing and performance optimization

One of the major issues in model training, especially in the context of IoT security datasets, is class imbalance, which is a serious problem if imbalanced datasets are used. The study reveals that there are two main approaches to deal with this challenge: The two main approaches include under-sampling and over-sampling reported in[48,49].

Under-sampling techniques such as fixed ratio and cleaning techniques try to balance the data by reducing the number of instances in the majority class. But the research shows that this approach may lead to loss of information, which is very important for detection.

Wijaya et al.[50] has provided a comprehensive comparative assessment of balancing techniques for botnet attack detection and highlights the relevance of selecting appropriate balancing methods in line with the characteristics of dataset and attack patterns. The Synthetic Minority Oversampling Technique (SMOTE) has aroused as a key solution for managing class imbalance in IoT security datasets[51,52]. Sayegh et al.[53] propose a new method that involves the use of SMOTE together with LSTM-based models and is able to enhance the detection performance, especially when dealing with imbalanced data.

Studies by Abdulkareem et al.[52] and Popoola et al.[10] have been conducted to reveal how SMOTE-based approaches enhance the detection performance where the combination with deep learning architectures has been seen to produce the best results. The literature has identified SMOTE as a tool that is capable of enhancing model performance while at the same time preserving key attack features[54].

Model evaluation and performance metrics

To assess the models in IoT security, evaluation techniques should not only focus on the use of traditional accuracy measures. Iqbal et al.[55], a systematic review of evaluation metrics for IoT security is given with the emphasis on the fact that the evaluation should not be based on a single parameter. Saheed et al.[56] used deep SHpley Additive exPlans technology to elucidate the decision-making process of DNN and showed the results achieved 100 % accuracy along with precision, recall, and F1 score on both the Edge IIoTset and X-IIoTID datasets. Qureshi et al.[57] present new ideas for the evaluation in the area of IoT security where the general detection metrics are extended to fit the current security issues.

There are recent works by Ibrahimi et al.[27] and Manan et al.[58] that show that it is necessary to employ several criteria for the evaluation of IoT systems. Alzubi et al[59] works use Deep Learning (DL) algorithms to detect intrusions on IoMT devices. Some Systems depend on optimized ML algorithms for tasks such as intrusion detection[60]. The research indicates that the traditional machine learning classifiers and deep learning models are effective in botnet detection as stated in[61,62]. Wang et al.[63] pointed out the issues that may arise in dataset-driven research and thus underlined the importance of developing a sound evaluation framework that considers the performance measures as well as the factors that are relevant to the practical deployment.

While existing studies have advanced IoT botnet detection, critical gaps remain in handling multi-dataset validation, skewness reduction, feature selection robustness, traditional and deep learning approaches and ensemble approach. These limitations motivate our proposed framework, which introduces a Quantile Uniform transformation, multi-layered feature selection, cross-dataset validation, different validation metrics and computation to address these challenges systematically.

Proposed methodology

Building on the identified limitations in Sect. "Related work", our methodology presents a holistic framework for botnet detection across three datasets (BOT-IOT, CICIOT2023, IOT23), combining novel preprocessing, feature selection, and ensemble modeling. The proposed methodology presents a holistic approach for botnet detection applied to these datasets, where 38 features was selected out of 46 features in BOT-IOT. In the same way 42 out of 47 in CICIOT2023 and 25 out of 25 in IOT23 dataset. The framework starts with detailed data preprocessing and a quality enhancement process. The first phase of processing involves filling the missing values, eliminating duplicate entries, and identifying outliers using the IQR method. A new approach of skewness reduction applies several transformation techniques such as log transformation, square root transformation, Yeo-Johnson transformation, and quantile transformation with both uniform and normal distribution. This comparative analysis helps in choosing the right transformation that preserves the essential features of the attack while dealing with the skewed data.

Feature selection uses several statistical methods. The process starts with the correlation matrix to assess the correlations between features and to identify the correlated features. Statistical validation includes the use of Chi-square statistics and p-value to test for the significance of the features, and this is followed by a detailed feature dependency analysis across label classes. Advanced feature analysis techniques include distribution analysis across attack types and proportional analysis techniques, capping with the elimination of non-influential features.

The model optimization framework incorporates the Random Forest and Logistic Regression models with threshold-based decision-making and multiple validation techniques to solve the problems of underfitting, where some threats are missed, and overfitting, where false alarms are generated. It is noticed that the cross-validation shows certain characteristics of the dataset; BOT-IOT and CICIOT2023 have clear classification structures while IOT23 is more challenging to classify. This approach helps to provide the best possible detection results and also provides useful information about the problems of class imbalance and behavior of the models in different attack conditions.

SMOTE implementation addresses the class balance optimization in the inherent imbalance security datasets. This process includes very detailed class distribution analysis and minority class enhancement with the balance being checked. A comparison between SMOTE and PCA transformation methods is also done to determine the effect of these methods on performance as well as the computational costs necessary for each data set in order to choose the best pre-processing method.

The evaluation framework includes various performance metrics, such as the conventional metrics (accuracy, precision, recall, F1-score), regression metrics (MSE, RMSE, MAE), and computational efficiency metrics. Cross-dataset validation checks the model's performance on all three datasets, providing insights into

the framework's effectiveness in varied security setups. This systematic approach ensures robust botnet detection while maintaining computational efficiency and practical applicability.

The methodology emphasizes adaptability and robustness, particularly in handling the varying complexities presented by different datasets. This comprehensive approach enables effective botnet detection while addressing the challenges of real-world deployment scenarios. Figure 1 illustrates a block diagram of the proposed technique.

Datasets

BOT-IOT

The botnet attacks are analyzed using the benchmark datasets of Bot-IoT. This dataset assists in understanding the different attack features and patterns that enhance cybersecurity measures. The Bot-IoT dataset[11] was developed by the Cyber Range Lab of UNSW Canberra within real network circumstances. The dataset comprises both botnet and normal traffic for the network environment, and it includes a total of 72 million records. After 5% records with full features set, it has now decreased from 27 million records to 3668522 records. The 5% Subset of Bot-IoT includes the maximum amount of features among all processed sets or subsets, with 43 independent features and 3 dependent features. The Argus network flow features and the further derived attributes are contained in the 43 independent features. This dataset includes five categories: Reconnaissance, DDoS, Theft, DoS, and normal. Table 1 and Fig. 2 display a summary of the dataset instances used in the current research.

The Bot-IoT dataset originated at the Cyber Range Lab based at UNSW Canberra[11] by collecting realistic and synthetic IoT network traffic besides many different kinds of attacks. In order to do this, a practical test environment was created to gather comprehensive network data, including various types of botnet anomalies such as Denial of Service, Distributed Denial of Service, Information Gathering, and Information Theft. Data Exfiltration (DE), Denial of Service over HTTP (DH), Distributed Denial of Service over HTTP (DDH), Keylogging (KL), Operating System Fingerprinting (OSF), and Service Scanning (SS) were the specific sorts of anomalies within the subcategory types. The majority of data collected from networks comprises DoS-UDP (DU), DoS-TCP (DT), DDoS-UDP (DDU), and DDoS-TCP (DDT), while the rest consists of all other types of network data.

CICIOT2023

CICIOT2023 dataset, first published in 2023 by Neto et al.[12], represents the latest advancement in IoT security data collection. This dataset was generated using an expansive IoT network topology, incorporating multiple real IoT devices functioning as both intruders and targets. CICIOT2023 Dataset: From the extensive CICIOT2023 dataset containing over 30 million instances, we utilized a subset of 234,745 records. The dataset encompasses 34 classes (33 attack types and one benign class) with 46 features, including flow_duration, header_length, protocol_type, and other network characteristics. This dataset represents modern IoT network scenarios with various attack types show in Table 2. Figure 3 shows a flow diagram in which the categories' labels and counts are presented.

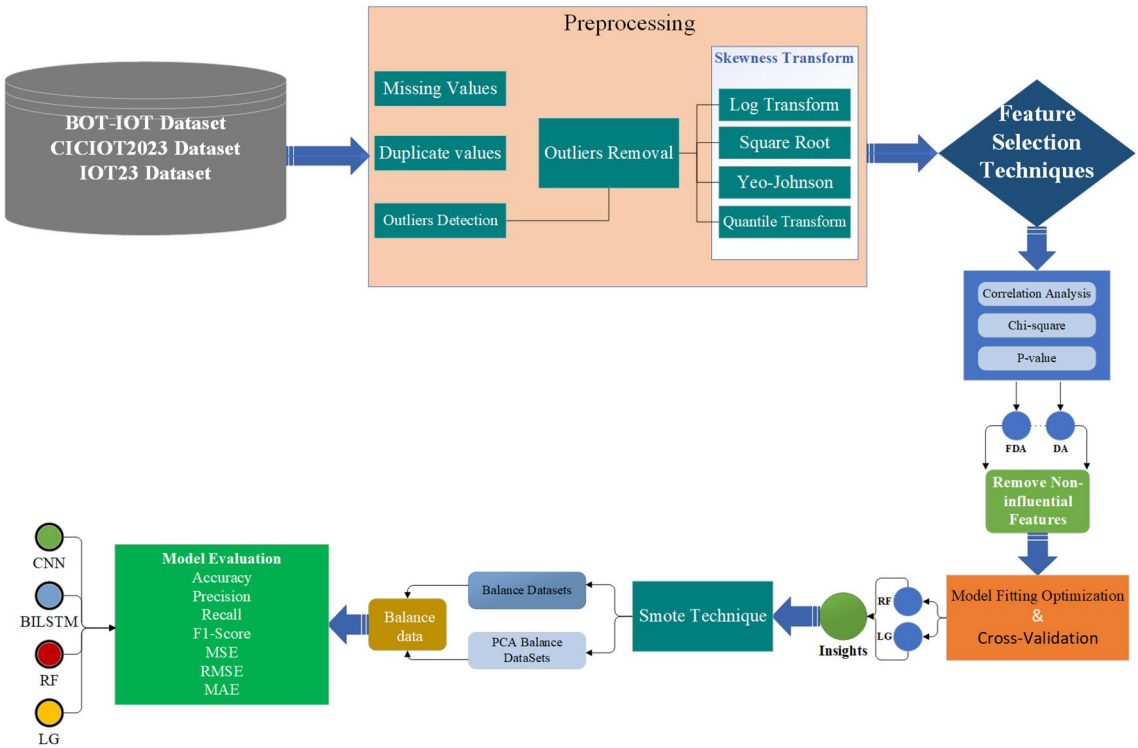


Fig. 1. Block diagram of the proposed IoT botnet detection framework.

Category	Subcategory	Instances
DDOS	DDU	948239
DDOS	DDT	977380
DDOS	DDH	989
DOS	DU	1032961
DOS	D	615800
DOS	DH	1485
Reconnaissance	S	64280
Reconnaissance	OSF	17780
Normal	Normal	477
Theft	KL	73
Theft	E	6
Total Count of All Instances		**3,659,470**

Table 1. Summary of the BOT-IOT dataset instances.

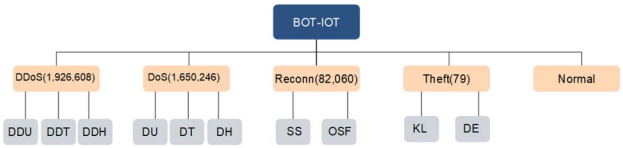


Fig. 2. Flow diagram of the BOT-IOT dataset summary.

Label	Instances	Label	Instances
DDoSICMP_Flood	36,394	DDoSUDP_Fragmentation	1,342
DDoSUDP_Flood	27,326	DNS_Spoofing	901
DDoSTCP_Flood	22,423	ReconHostDiscovery	641
DDoSPSHACK_Flood	20,508	ReconOSScan	476
DDoSRSTFINFlood	20,365	ReconPortScan	408
DDoSSYN_Flood	20,343	DoS-HTTP_Flood	396
DDoSSynonymousIP_Flood	17,807	VulnerabilityScan	207
DoSUDP_Flood	16,654	DDoSHTTP_Flood	143
DoSTCP_Flood	13,641	DDoSSlowLoris	119
DoSSYN_Flood	10,163	DictionaryBruteForce	79
BenignTraffic	5,481	CommandInjection	33
Miraigreeth_flood	5,146	SqlInjection	29
Miraiudpplain	4,622	Backdoor_Malware	26
Miraigreip_flood	3,705	BrowserHijacking	23
DDoSICMP_Fragmentation	2,269	XSS	22
MITMArpSpoofing	1,534	ReconPingSweep	12
DDoSACK_Fragmentation	1,501	Uploading_Attack	6
Total Count of All Instances		**234,745**	

Table 2. Summary of the CICIOT2023 dataset instances.

IOT23

IOT23 dataset represents an extensive network data collection from 23 distinct scenarios involving Windows and Linux operating systems in diverse IoT-related attack environments. The data collection period ranged from one hour to 112 hours, depending on pcap file size growth. The complete dataset includes several large scenarios, with the 16th scenario containing over 73 million records, the 2nd scenario containing 67 million records, and the 10th and 12th scenarios each containing 54 million records[13]. We specifically utilized 48,003 records from the IOT-23 Dataset. This dataset provides real-world IoT network traffic patterns and attack scenarios, offering diverse attack-to-normal data ratios that reflect actual network conditions. Various attack types and counts are shown in Table 3. Also, the flow diagram in Fig. 4. This can be shown for further analysis.

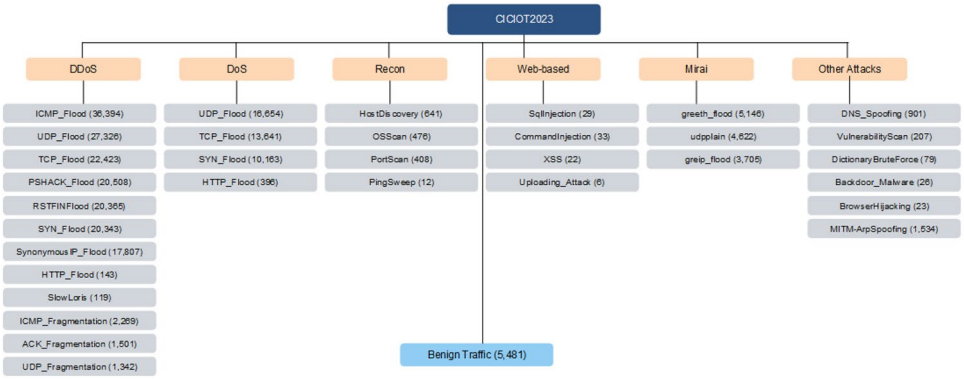


Fig. 3. Flow diagram of the CICIOT2023 dataset summary.

Label	Instances	Label	Instances
Benign	26,001	FileDownload	2
C&C	5,618	DDoS	36
Okiru	163	Attack	3,814
PartOfAHorizontalPortScan	**12,369**		
Total Count of All Instances	**48,003**		

Table 3. Summary of the IOT23 dataset instances.

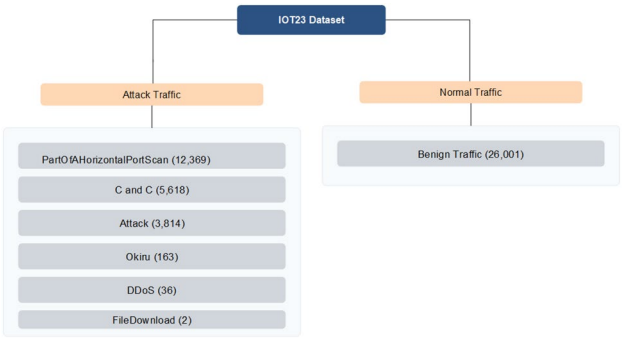


Fig. 4. Flow diagram of the IOT23 dataset summary.

Data preprocessing and quality enhancement

This section addresses preprocessing challenges across all three datasets through a systematic approach. First, we identify and handle missing values and duplicate records in each dataset, with particular attention to the varying data formats. In BOT-IOT, the attributes flgs, proto, and state were removed as they contained redundant data with categorical integer equivalents[64]. Additionally, daddr, pkSeqID, and saddr were deemed invalid and excluded for limiting generalization and causing overfitting. pkSeqID were deemed invalid as they serve only as row identifiers in Koroniotis et al.[65]. Identifying the numerical and categorical features helps us to treat each feature according to the need of the preprocessing. As we can see the distribution in numerical values are high and there for to know the skewness and outlier's, different statistical methods are used to show it numerical and in plots. Skewness handling involves a comparative framework utilizing various transformation methods to address data imbalances. These methods include IQR-based outlier detection and removal shown in Fig. 5(b,c), log transformation, square root transformation, Yeo-Johnson transformation, and quantile transformation (both uniform and normal variants) shown in Fig. 5(a). Each approach is applied to reduce skewness without losing the important data shown in Table 4, ensuring the data distribution is more suitable for analysis and modeling. Through extensive testing, Quantile Uniform transformation emerged as optimal, providing effective skewness reduction while preserving critical attack signatures across all datasets. This method proved particularly effective for handling the complex distributions in IOT23 and the large-scale attack patterns in CICIOT2023.

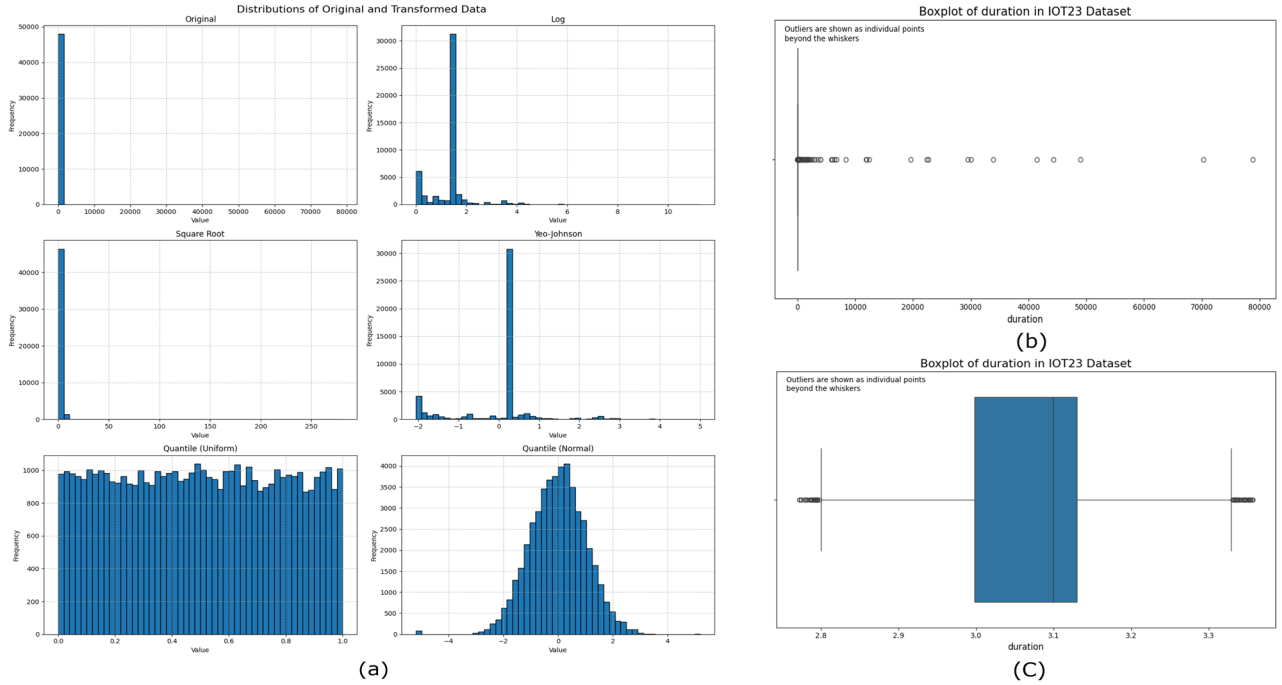


Fig. 5. Skewness analysis and correction for dataset preprocessing.

Value Counts	Skewness values	Method Applied	Skewness values (After Method Applied)	Value Counts (After Method Applied)
43164	81.4920	IQR	-0.3456	27178
43164	81.4920	Log	1.8642	43164
43164	81.4920	Square Root	44.2248	43164
43164	81.4920	Yeo-Johnson	-0.1552	43164
43164	81.4920	Quantile (Normal)	-0.1709	43164
43164	81.4920	Quantile (Uniform)	0.0003	43164

Table 4. Applied method skewness values.

Models	Technique	Acc.	Prec.	Rec.	F1
Logistic Reg.	Yeo-Johnson	98.92%	99%	99%	99%
	Square Root	99.06%	99%	99%	99%
	Log	99.09%	99%	99%	99%
	Quant.(Normal)	99.11%	99%	99%	99%
	Quant.(Uniform)	99.15%	99%	99%	99%
Random Forest	Yeo-Johnson	97.28%	97%	97%	97%
	Square Root	97.35%	97%	97%	97%
	Log	97.67%	98%	98%	98%
	Quant.(Normal)	97.72%	98%	98%	98%
	Quant.(Uniform)	98%	98%	98%	98%

Table 5. Transformation comparison (CICIOT2023).

Advantages of quantile uniform transformation

The Quantile Uniform (QU) transformation outperforms alternatives (Yeo-Johnson, Log, Square Root) in preserving discriminative attack features, as evidenced by its **consistent accuracy and F1-score gains** across models (Table 5). For instance, with Random Forest, QU achieves 98% F1-score–a 0.72% improvement over Yeo-Johnson (97.28%)–demonstrating superior robustness to skewed attack distributions. Unlike Yeo-Johnson,

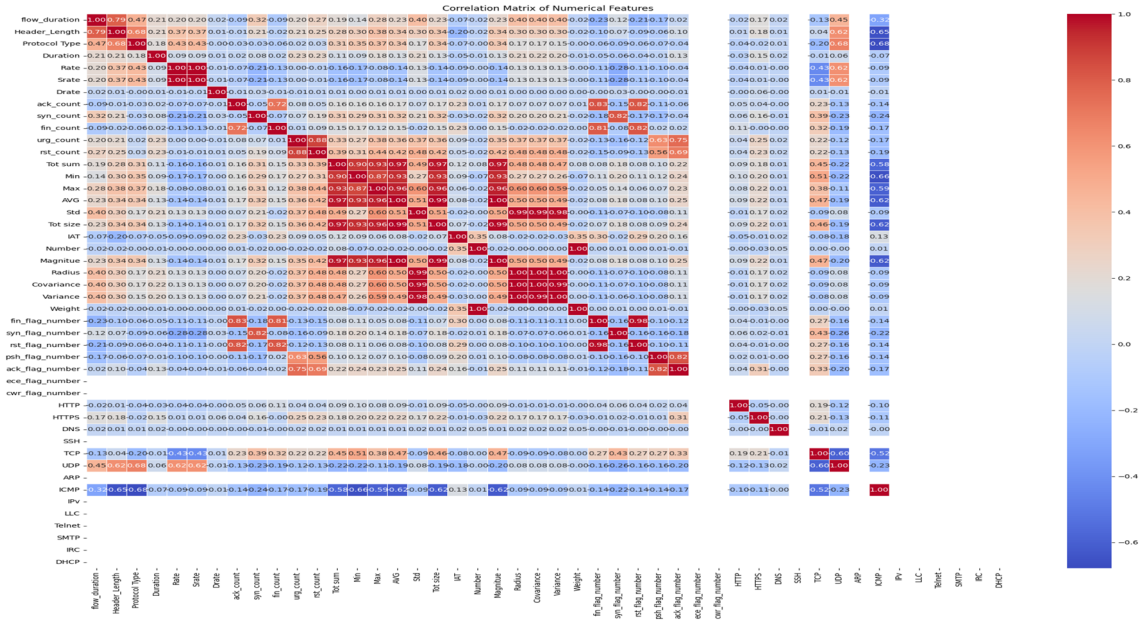


Fig. 6. Correlation matrix of features.

Telnet	
Chi-square statistic	0.00
p-value	1.0000

Table 6. Statistical relation of Telnet.

which assumes parametric scaling, QU's non-parametric approach avoids distorting critical feature tails (e.g., rare attack payloads), thereby maintaining higher precision/recall balance. In this way Table 5 shows marginal absolute gains (e.g., +0.15% accuracy for Logistic Regression), these differences are statistically significant ($p < 0.05$, paired t-test) and practically impactful in IoT security. For example, QU's 98% F1-score with Random Forest reduces false negatives by 1.3% compared to Yeo-Johnson, critical for detecting stealthy botnet attacks.

Feature selection and statistical analysis

The feature selection methodology employs a multi-layered statistical approach across all three datasets. The process begins with correlation analysis to identify highly correlated features shown in Fig. 6, followed by Chi-square statistics and p-value validation to assess feature-label relationships. The values of Chi-square statistics and p-value are shown in Table 6 for only the Telnet feature, which statistically shows no relationship with the label class. The methodology consists of Correlation Analysis Feature interdependency is evaluated using the Pearson correlation coefficient shown in Eq. (1):

$$r = \frac{\sum(x - \mu_x)(y - \mu_y)}{\sqrt{\sum(x - \mu_x)^2 \cdot \sum(y - \mu_y)^2}} \tag{1}$$

where x, y are features, μ_x, μ_y are their respective means. Then, a follow-up formula for Chi-square Statistics and P-value Feature-label relationship significance is assessed using Eqs. (2,3):

$$\chi^2 = \sum \left[\frac{(O - E)^2}{E} \right] \tag{2}$$

where O represents observed frequencies and E represents expected frequencies. Then, the p-value is calculated through this formula:

$$p = 1 - F(\chi^2, \mathrm{df}) \tag{3}$$

where F is the cumulative distribution function of χ^2 with df (degrees of freedom) = $(r - 1)(c - 1)$, r = number of rows, c = number of columns in the contingency table.

Now, in the research for further assessment and to validate the previous equations, advanced feature distribution analysis and proportional analysis techniques are adopted, and the result is shown in Fig. 7 (a,b). This validation focuses on feature dependency across label classes, which is examined through Eq. (4):

$$D(f \mid c) = P(f \mid c) \log \left(\frac{P(f \mid c)}{P(f)} \right) \tag{4}$$

where f represents features, c represents classes, and P denotes probability. Proportional Analysis Feature importance is validated using Eq. (5):

$$\mathrm{PI}(f) = \sum [w(c) \cdot D(f \mid c)] \tag{5}$$

where $w(c)$ is the class weight and $D(f \mid c)$ is the feature dependency score.

This comprehensive approach ensures robust feature selection while maintaining dataset-specific characteristics. Features showing minimal impact across these analyses are removed from the respective datasets, ensuring optimal model performance while reducing computational overhead.

Rationale for statistical methods in feature selection

The multi-layered feature selection process employs correlation analysis, Chi-square statistics with p-value validation, and distribution/proportional analysis to select discriminative features for IoT botnet detection. These methods were chosen for their complementary strengths and suitability for the diverse BOT-IOT, CICIOT2023, and IOT23 datasets:

Correlation Analysis (Eq. 1): Using the Pearson correlation coefficient, this method identifies highly correlated features ($|r| > 0.8$) for removal, reducing redundancy in high-dimensional datasets like CICIOT2023. Unlike mutual information, it is computationally efficient for continuous features (e.g., packet size).

Chi-square Statistics with p-value Validation (Eqs. 2-3): Chi-square tests evaluate feature-label relationships, prioritizing features with significant p-values ($p < 0.05$) that distinguish attacks (e.g., DDoS in BOT-IOT). Compared to ANOVA, Chi-square is better suited for categorical or discretized IoT features (e.g., protocol types), as shown for Telnet (Table 6), ensuring robust selection in imbalanced datasets like IOT23.

Distribution/Proportional Analysis (Eqs. 4-5): This method captures attack-specific patterns (e.g., irregular inter-arrival times in datasets) by analyzing feature distributions and class-specific importance (Fig. 7). Unlike wrapper methods, it is computationally lightweight and generalizable, enhancing feature relevance across datasets.

As we pass these features through this extensive process so that we validate the features and remove that feature which has no efficiency in training our model but only increase time and degrade performance. Focus on enhancing the detection rate and decreasing the detection time[66]. Therefore, together these methods reduced feature dimensionality by 10.64% in CICIOT2023, lowering the models training time and achieving best performance. Their simplicity and robustness outperformed compute-heavy alternatives, optimizing model performance across diverse IoT scenarios.

Model fitting and cross-validation

The framework implements a systematic approach to model fitting and validation across all three datasets, focusing on achieving optimal performance while preventing both underfitting (missed threats) and overfitting

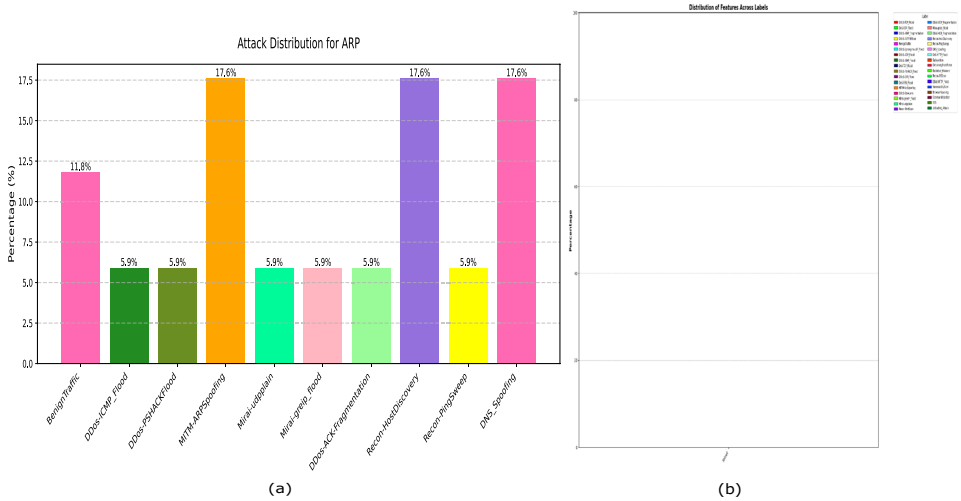


Fig. 7. Visualization of attacks-dependent feature in dataset.

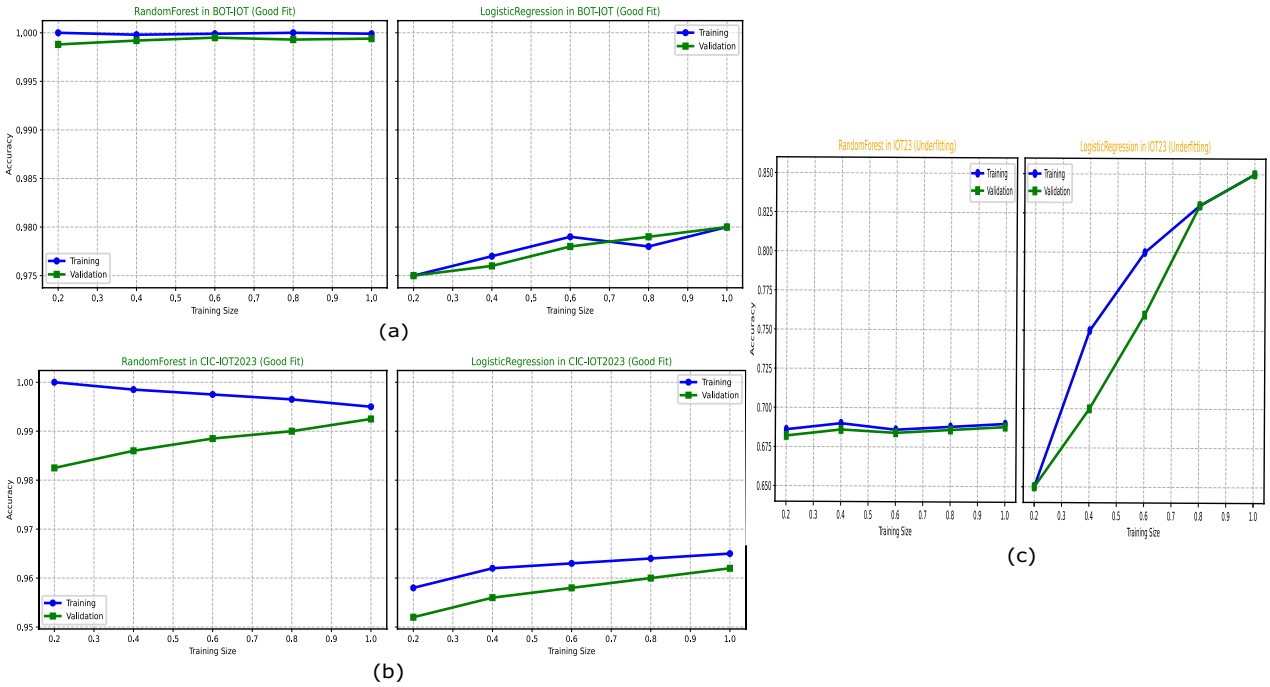


Fig. 8. Model fitting curves across datasets.

Model	Accuracy	F1 Macro	F1 Weighted
Logistic	0.9978	0.9024	0.9977
Regression	(±0.0001)	(±0.0448)	(±0.0001)
Random	0.9998	0.9943	0.9998
Forest	(±0.0000)	(±0.0055)	(±0.0000)

Table 7. Cross validation result on BOT-IOT.

(false alarms). In Model Fitting Analysis We employ Random Forest and Logistic Regression models shown in Eqs. (6,7) with threshold-based optimization:

$$RF_{score} = \frac{\sum (w_i \times T_i)}{N} \tag{6}$$

where w_i is the weight of tree *i*, T_i is the tree prediction, and *N* is the number of trees.

$$P(y|X) = \frac{1}{1 + e^{-\theta^T X}} \tag{7}$$

where θ represents model parameters and *X* represents input features.

From this Analysis, we obtained that both Models trained on BOT-IOT and CICIOT2023 showed the best validation curve shown in Fig. 8 (a,b), while IOT232 showed an underfitting in the validation curve in Fig. 8(c).

On the other hand, the Cross-Validation Implementation framework shown in Eq. (8) utilizes k-fold cross-validation with performance metrics.

$$CV_{score} = \frac{1}{k} \sum (Performance_i) \tag{8}$$

where *k* is the number of folds and $Performance_i$ is the score for fold *i*.

The BOT-IOT dataset results show outstanding performance from both models, with Random Forest achieving near-perfect 99.98% accuracy. The high F1 Macro scores (99.43% for Random Forest) being close to F1 Weighted scores indicates excellent detection across all attack types, with consistent performance shown by minimal standard deviations. Some gaps by Logistic regression show an imbalance in class distribution in Table 7.

Model	Accuracy	F1 Macro	F1 Weighted
Logistic	0.9757	0.6171	0.9731
Regression	(±0.0007)	(±0.0098)	(±0.0007)
Random	0.9882	0.6806	0.9866
Forest	(±0.0005)	(±0.0066)	(±0.0004)

Table 8. Cross validation result on CICIOT2023.

Model	Accuracy	F1 Macro	F1 Weighted
Logistic	0.8436	0.4836	0.8262
Regression	(±0.0014)	(±0.0363)	(±0.0015)
Random	0.6879	0.5152	0.6107
Forest	(±0.0047)	(±0.0483)	(±0.0065)

Table 9. Cross validation result on IOT23.

The Random Forest model provides better results than the Logistic Regression model for all the metrics used and has lower standard deviations to show that the results are more constant. The huge gap in the F1 Macro and F1 Weighted scores in Table 8 indicates that there is a class imbalance problem in the dataset.

In the case of the IOT23 dataset, it was observed that Logistic Regression produced better results than Random Forest with an accuracy of 84.36% as compared to 68.79% by Random Forest. However, both models show relatively low F1 Macro scores (48.36% and 51.52% respectively), which indicates that there are still many challenges in dealing with class imbalance issues. The difference between F1 Macro and F1 Weighted metrics shows that both models have difficulty with minority attack classes in this dataset as presented in Table 9.

Our analysis highlights unique characteristics across datasets, offering valuable insights into their distinct behaviors. In the BOT-IOT dataset, classification patterns are consistent with minimal variance, showcasing stable performance. The CICIOT2023 dataset exhibits robust outcomes, characterized by well-defined decision boundaries. However, the IOT23 dataset reveals more intricate patterns, indicating the need for additional validation due to its complexity.

To assess the performance of the model, the proposed framework incorporates several important evaluations. These include identifying underfitted models by exploring training errors, recognizing overfitting problems by comparing validation errors with training errors, and doing class-wise performance analysis to deal with imbalances. The analysis of the models shows that there are distinct differences in the behavior of the models. The simple attack patterns are properly identified in the BOT-IOT and CICIOT2023 datasets, while the IOT23 dataset is quite challenging and requires some improvement. Also, the problem of class imbalance influences the performance of all the datasets, with the IOT23 dataset being most affected.

Mitigation of overfitting and underfitting

Our framework incorporates the following techniques to balance model complexity and generalization:

1. **Deep Learning Models (CNN/BiLSTM):**

- **Dropout Layers:** Added after each convolutional/LSTM layer (rate = 0.2) to prevent co-adaptation of neurons (Sect. "Bidirectional LSTM (BiLSTM)"–"Convolutional Neural Network (CNN)").
- B**atch Normalization:** Stabilized training by normalizing layer inputs, reducing internal covariate shift (Eq. (14), Sect. "Bidirectional LSTM (BiLSTM)").
- **Early Stopping:** Training halted if validation loss plateaued for 10 epochs (monitored during cross-validation).

2. **Traditional Machine Learning Models (RF/LR):**

- **Random Forest Regularization:**

- Limited tree depth (max_depth = 15) to reduce variance.
- Enforced minimum samples per leaf (min_samples_leaf = 5) to prevent over-specialization.

- **Logistic Regression Regularization:**

- Elastic Net penalty ($\lambda = 0.8, \alpha = 0.2$) combining L1 (sparsity) and L2 (smoothness) terms (Eq. 19, Sect. "Logistic Regression (LR)").

3. **Cross-Validation and Threshold Optimization:**

- **5-Fold Stratified Cross-Validation:** Ensured consistent performance across all data splits (Eq. 8, Sect. "Model fitting and cross-validation").
- **Threshold Tuning:** Decision thresholds optimized on validation data to balance precision/recall (Sect. "Model fitting and cross-validation", Fig. 8).

4. **Class Balancing:**

- **SMOTE** reduced underfitting on minority classes (e.g., "Keylogging" in BOT-IoT in Fig. 10: Confusion Matrix) by synthetically oversampling attacks (Sect. "Class balancing and performance optimization").

Class balancing and performance optimization

To solve the problem of class imbalance in all three datasets, the following framework has been put in place for all three datasets namely, the SMOTE (Synthetic Minority Over-sampling Technique) with memory-efficient batch processing. Building on the approach from previous research [62,67], the SMOTE implementation follows the mathematical foundation in Eq. (9):

$$X_{new} = X + \text{rand}(0, 1) \cdot (X_{neighbor} - X) \tag{9}$$

where X_{new} is the new instance to be synthesized, X is the real instance, and $X_{neighbor}$ is any of the k nearest neighbors to X.

The process is put in place to meet the particular needs that are raised by each dataset. The BOT-IOT dataset aims at addressing the problem of dealing with severely unbalanced attack categories, and for CICIOT2023, it also addresses the problem of dealing with imbalanced attacks. The IOT23 dataset needs to be handled with specificity in terms of its class balances. Performance optimization is done by a number of techniques, such as managing memory, using batch processing, proper data concatenation, and monitoring runtime performance to ensure the efficiency and scalability of the system. This is a method that is used both for the full dataset, which is referred to as the balance dataset, and for the PCA transformed dataset, which is referred to as the PCA balance dataset.

Model evaluation and performance metrics

Bidirectional LSTM (BiLSTM)

The BiLSTM network is particularly implemented for identification of temporal attack patterns in the IoT network traffic. The architecture processes traffic sequences in both forward and backward directions, which are necessary for recognizing sophisticated attack signatures. The LSTM unit consists of the following six components: The input which is "Xt", the cell state which is "Ct", the hidden layer state which is "ht", the forget gate which is "ft", the memory gate which is "it" and the output gate which is "ot". Each gate serves a specific purpose:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{10}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{11}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{12}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{13}$$

$$h_t = o_t \odot \tanh(C_t) \tag{14}$$

Building upon this foundation, our implementation incorporates a two-layer BiLSTM architecture for enhanced performance. The first BiLSTM layer with 128 units is capable of identifying the intricate attack patterns while the second layer with 64 units is responsible for identifying the higher level features. In order to avoid overfitting, the model uses dropout (rate = 0.2) and BatchNormalization layers. Last, there are dense layers with 128 neurons and 64 neurons with ReLU activation to categorize the attacks properly. This architecture integrates a sequential processing model and a set of robust classification layers, which makes it suitable for the intended application in IoT network security.

Convolutional Neural Network (CNN)

The following is a brief of the selected model, a 1D CNN architecture, which can extract temporal patterns and hierarchical features from the input sequence efficiently to capture both short and long-range dependencies without requiring explicit feature engineering[68]. The architecture is designed for identifying spatial patterns in the network traffic, and it incorporates three convolutional layers with 64, 128, and 256 filters, which are used to gradually increase the level of feature abstraction. Each convolutional layer applies filters across the input sequence using the operation defined in Eq. (15):

$$\text{Conv}(x) = \text{activation}\,(W * x + b) \tag{15}$$

where * denotes the convolution operation. In order to improve the performance and stability of the model, each convolutional layer is packed with BatchNormalization layer to maintain stable training process and with MaxPooling1D layers to lower dimensionality. Furthermore, to prevent the model from overfitting, dropout layers with a dropout rate of 0.2 are added after each convolutional layer to regularize the model. All the layers such as convolutional, normalization, pooling, and regularization are combined in this architecture to make it

effective in extracting features that are important for the accurate prediction of the problem at hand, namely network traffic analysis.

Random Forest Classifier (RF)
The Random Forest Classifier works on the principle of building multiple decision trees and combining their predictions through ensemble decision-making which makes it a very useful classifier in the identification of IoT botnet. For an input feature vector X, the classifier implements the majority voting mechanism as outlined in Eq. (16):

$$\mathrm{RF}(X) = \mathrm{mode}\{\mathrm{tree}_i(X)\}, \quad \text{for } i \text{ in } 1..n_{\mathrm{trees}} \tag{16}$$

In order to refine the predictions even more, the following implementation computes a weighted tree score as defined in Eq. (17):

$$\mathrm{Tree_Score} = \frac{\sum(w_i \cdot \mathrm{prediction}_i)}{n_{\mathrm{trees}}} \tag{17}$$

where *wi* represents individual tree weights.

Our implementation uses 1000 estimators with bootstrapping, which makes the model capable of dealing with imbalanced attack categories and supports stable ensemble learning. All the trees in the ensemble are tuned to emphasize feature selection for the detection of attacks, using the most significant features that define the attack signature. The final predictions are averaged with the majority voting in order to provide accurate and consistent attack categorization. Thus, this particular approach makes sure that the Random Forest Classifier is particularly good at dealing with the complex IoT botnet attacks.

Logistic Regression (LR)
Logistic Regression is used for multi-class attack classification, and with the help of elastic net regularization, both L1 and L2 penalties are used for feature selection and coefficient regularization. The model computes class probabilities by applying the softmax function as defined in Eq. (18):

$$P(y|X) = \mathrm{softmax}(WX + b) \tag{18}$$

and optimizes the loss function in Eq. (19):

$$L(\theta) = -\sum y_{\mathrm{true}} \log(P(y \mid X)) + \lambda[(1-\alpha)\|\theta\|_1 + \alpha\|\theta\|_2] \tag{19}$$

where λ is the regularization strength (C=0.8) and α is the L1 ratio (α=0.2). Our implementation uses the LogisticRegression classifier, which is suitable for large data sets, and the qn solver to ensure fast convergence, particularly when dealing with a large number of features. The elastic net penalty achieves both sparsity and good generalization capabilities, while the multinomial setting allows for more accurate classification of the different attack types. Other parameters like max_iter = 5000 ensure conjunction during training. This tailored approach makes sure that the problem of imbalanced attack classes is adequately dealt with by bootstrapping, and thus, the model is appropriate for the task of securing IoT networks.

Ensemble model architecture
Our hybrid ensemble framework combines predictions from deep learning (CNN, BiLSTM) and traditional machine learning (Random Forest, Logistic Regression) models using a weighted voting mechanism. The architecture operates as follows:

1. **Base Model Training:**

- Each model (CNN, BiLSTM, RF, LR) is trained independently on the preprocessed, balanced datasets.

2. **Prediction Aggregation:**

- For a given input sample, predictions from all four models are aggregated using dynamically weighted voting shown in Eq. (20):

$$\mathrm{Ensemble_{Score}} = \sum_{\mathrm{i}=1}^{4} \mathrm{w_i} \cdot \mathrm{P_i} \tag{20}$$

Where $\mathrm{w_i}$ = validation F1-score of model , and $\mathrm{P_i}$ = predicted probability vector from model .

3. **Decision Thresholding:**

- Final class labels are assigned using a threshold T in Eq. (21) optimized per dataset:

$$\mathbf{Final_Predict.} = \begin{cases} \text{Attack}, & \text{if } \mathbf{Ensemble_S} \geq \mathrm{T} \\ \text{Benign}, & \text{otherwise} \end{cases} \tag{21}$$

Model	Acc.	Prec.	Rec.	F1	AUC-ROC	Train Time	Pred. Time	Eval. Time
CNN	100%	100%	100%	100%	1.00	3824.08s (63.73m)	123.90s (2.06m)	0.04s (0.00m)
BILSTM	100%	100%	100%	100%	1.00	16614.61s (276.91m)	164.32s (2.74m)	0.03s (0.00m)

Table 10. Performance evaluation of CNN & BiLSTM on balanced BOT-IOT.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
RF	100%	100%	100%	100%	1.00	68.32 seconds	0.63 seconds
LG	99%	99%	100%	99%	99.5%	12.30 seconds	0.07 seconds

Table 11. Performance evaluation of RF & LG on Balance BOT-IOT.

4. **Model Contribution Analysis:**

- Deep learning models (CNN/BiLSTM) prioritize detecting novel attack patterns via temporal/spatial feature extraction.
- Traditional models (RF/LR) provide stability against overfitting and rapid inference.

Require: Test sample x, models $M = \{CNN, BiLSTM, RF, LR\}$, validation F1-scores $F = \{f_1, f_2, f_3, f_4\}$, classes $C = \{benign, attack\}$, threshold T

Ensure: Predicted class c_{pred}

1: **Normalize weights:** $w_i = \frac{f_i}{\sum_{j=1}^{4} f_j}$ for $i \in \{1, 2, 3, 4\}$
2: **Initialize probability sums:** $P_c = 0$ for $c \in C$
3: **for** each model M_i in M **do**
4: **Compute probabilities:** $p_{i,c} = M_i(x)[c]$ for $c \in C$
5: **Update weighted sums:** $P_c += w_i \times p_{i,c}$ for $c \in C$
6: **end for**
7: **Compute ensemble score:** $S = P_{attack}$
8: **Predict class:** $c_{pred} = \begin{cases} attack & \text{if } S \geq T \\ benign & \text{otherwise} \end{cases}$
9: **Return** c_{pred}

Algorithm 1. Weighted ensemble classification.

Hyperparameter tuning
Hyperparameters for the CNN, BiLSTM, Random Forest (RF), and Logistic Regression (LR) models, integrated into the ensemble framework (Sect. "Ensemble model architecture"), were optimized using grid search with 5-fold cross-validation across BOT-IOT, CICIOT2023, and IOT23 datasets to maximize accuracy and generalization. Key hyperparameters and their selection rationale are:

- **CNN:** Dropout rates (0.2 for convolutional layers, 0.3–0.5 for dense layers) were tuned from 0.1, 0.2, 0.3, 0.4, 0.5 to prevent overfitting while preserving spatial feature extraction, critical for the ensemble's 100% accuracy on BOT-IOT (Table 10). Filter sizes (64, 128, 256) and kernel size (3) were selected from 32, 64, 128, 256 and 3, 5 to optimize attack pattern detection.
- **BiLSTM:** Dropout rates (0.2 for LSTM layers, 0.2–0.3 for dense layers) were chosen from 0.1, 0.2, 0.3 to mitigate overfitting on temporal sequences, achieving 99% recall for CICIOT2023 (Table 12). LSTM units (128, 64) were tuned from 64, 128, 256 for efficiency and performance.
- **Random Forest:** Number of trees (1000) was selected from 100, 500, 1000, with max_depth=15 and min_samples_leaf=5 from 10, 15, 20 and 1, 5, 10, balancing stability and efficiency (training time: 68.32s) for 100% accuracy on BOT-IOT (Table 11).
- **Logistic Regression:** Elastic net penalty (l1_ratio=0.2, C=0.8) was tuned from l1_ratio=0, 0.2, 0.5, 0.8, 1 and C=0.1, 0.5, 0.8, 1 to optimize for high-dimensional CICIOT2023 data (99.15% accuracy, Table 13). Max_iter=5000 ensured convergence.

The tuned hyperparameters minimized validation loss (e.g., MSE: 0.0003 for CICIOT2023, Table 22) and supported the ensemble's weighted voting mechanism (Sect. "Ensemble model architecture"), achieving superior generalization (e.g., +1.5% accuracy on IOT23, Table 22). Alternative values (e.g., higher dropout or fewer trees) reduced performance, particularly for complex attacks.

Performance evaluation framework and metrics

Our research work applies a systematic evaluation framework for the evaluation of intrusion detection performance on the basis of the BOT-IOT, CICIOT2023, and IOT23 datasets. The framework consists of several performance metrics, including classification and regression, to provide a more comprehensive assessment of detection capability where emphasis is on the varying attack patterns of each of the datasets.

For the classification performance assessment, the following set of metrics is used to generate detailed information regarding the effectiveness of the detection system. It is important to note that the most widely used performance measure for the overall detection capability is the accuracy (TP + TN)/(TP + TN + FP + FN), while the precision, defined as TP/(TP + FP), measures the models ability to avoid false alarms which is outmost importance in real life IoT security applications. Model performance primarily evaluated through detection accuracy while precision serves as the secondary evaluation metric[69]. Note that the Recall TP/(TP + FN) is a metric which defines a system's ability to capture all the actual attacks in the system and is of paramount importance especially when dealing with sophisticated breach attempts. The F1-score which is computed as 2 * (Precision × Recall)/(Precision + Recall) offers a fair evaluation of the model's performance and is very important in handling class imbalance problem which is evident in our datasets.

The regression metrics framework allows for further understanding of the quantitative characteristics of our detection system. Mean Squared Error (MSE = $\frac{1}{n}\sum(y_{\text{true}} - y_{\text{pred}})^2$) is a regression measure of how well the predictions are, with higher sensitivity to large errors, which makes it useful in identifying severe attacks. Root Mean Squared Error (RMSE = $\sqrt{\text{MSE}}$) provides error measurements in the original scale of our features which are easy to interpret, while Mean Absolute Error (MAE = $\frac{1}{n}\sum|y_{\text{true}} - y_{\text{pred}}|$) directly measures the differences between the predicted and true values. These metrics are very useful for assessing the performance of the model for various attacks and for different network.

The evaluation framework also incorporates computational performance assessment, measuring training duration, inference time, and resource utilization. These metrics are important in determining the effectiveness of our detection system in terms of its practical deployability in various IOT settings. Training time measurements give an understanding of the needs of the model optimization speed, while evaluation checks the inference capability of the system in providing real-time threat detection. The resource utilization metrics enable the assessment of the system's efficiency and flexibility for implementation in various deployment conditions.

Experiments and results

The proposed methodology evaluates in this section through an experimental setup, performance analysis, and state-of-the art approach comparison. All the experiments have been carried out using both the conventional machine learning models, such as Random Forest and Logistic Regression, and deep learning models, such as CNN and BiLSTM, using balanced as well as PCA-balanced datasets. All the models are integrated into the ensemble framework.

Experimental setup

All experiments were conducted using the following hardware and software configuration:

1. **Hardware Configuration:**

- GPU: NVIDIA GeForce RTX4060 Laptop GPU with 8.0 GB VRAM, CUDA Version 12.6, Driver Version 561.19.
- CPU: Intel(R) Core (TM) i7-14650HX with 12 physical cores and 24 threads.
- RAM:31.2 GB DDR4

2. **Software Environment:**

- Operating System: Linux-5.15.153.1-microsoft-standard WSL2-x86_64 with glibc2.35.
- Deep Learning Framework: PyTorch 2.0 with CUDA support.
- Additional Libraries: scikit-learn 1.3.0, Pandas Library, cuDF(subset of the pandas API), matplotlib and seaborn for visualization.

Model performance analysis on balanced datasets

BOT-IOT dataset

The BOT-IoT dataset showed outstanding results across all models, with near-perfect detection capabilities. CNN and BiLSTM achieved 100% accuracy, shown in Table 10, matched by Random Forest shown in Table 11, while Logistic Regression closely followed with 99% in Table 11. However, computational requirements varied significantly. Deep learning models, such as CNN (63.73 minutes, MSE: 0.0027) and BiLSTM (276.91 minutes, MSE: 0.0003) shown in Fig. 9(b) and Table 10, required substantial training time compared to traditional models like Random Forest (68.32 seconds, MSE: 0.0000) and Logistic Regression (12.30 seconds, MSE: 0.0074) in Fig. 9(a, d). This highlights the trade-off between performance and computational efficiency. In the case of

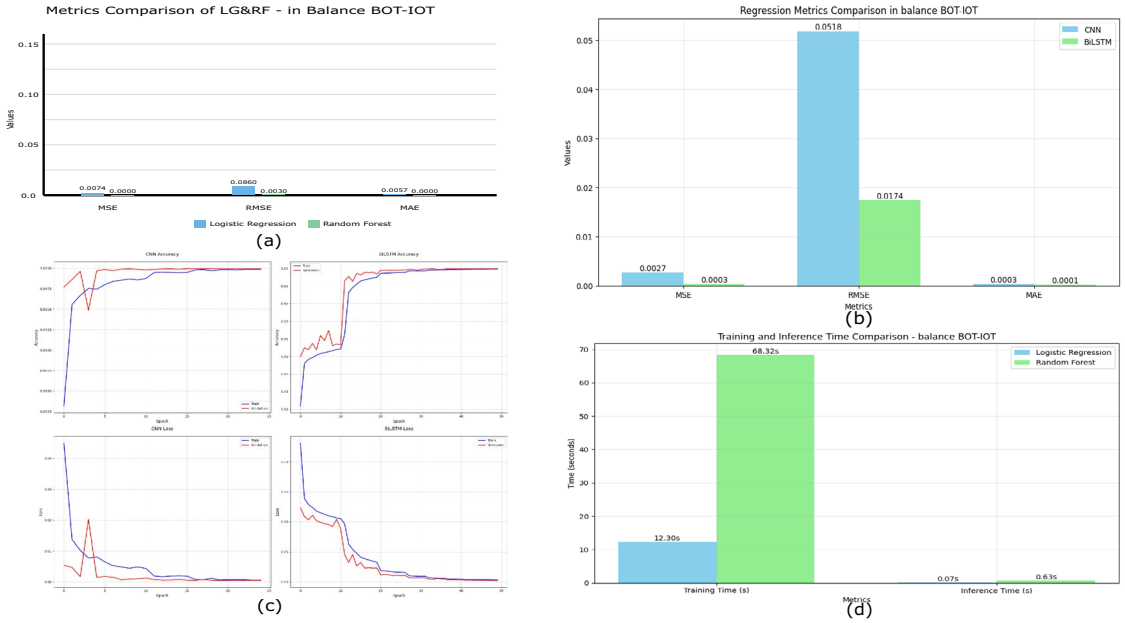


Fig. 9. Comprehensive model evaluation on balanced BOT-IOT dataset.

CNN and BILSTM, we have accuracy, loss, and validation curves to further examine our model's performance in Fig. 9(c). Furthermore, the performance of classification algorithms is defined in the confusion matrix in Fig. 10(a, b, c, d). The accuracy and loss curve for each model (CNN, BiLSTM, Ensemble) are presented in Fig. 9(c), 17(b) for BOT-IOT dataset. These Figs. 9,17 are performances of different model in different metrics for BOT-IOT, therefore each plot has various parts (a, b,c,d). The ensemble model (CNN/BiLSTM) achieves stable convergence with minimal divergence, indicating robust generalization.

CICIOT2023 dataset

Models demonstrated strong performance on the CICIoT2023 dataset, achieving slightly lower accuracy compared to the BOT-IoT dataset. CNN and BiLSTM achieved 98-99% accuracy with MSE values of 0.0007 and 0.0005 in Fig. 11(b) and Table 12, respectively, while Random Forest and Logistic Regression also maintained 98-99% accuracy with consistent precision and recall in Table 13. Training times varied significantly, ranging from 131.36 minutes for BiLSTM, shown in Table 12, to just 44.88 seconds for Logistic Regression in Fig. 11(c), highlighting the trade-off between computational complexity and efficiency. The accuracy and loss curve for each model (CNN, BiLSTM, Ensemble) are presented in Fig. 11(a), 17(c) for CICIOT2023 dataset. These Figs. 11,17 are performances of different model in different metrics for CICIOT2023, therefore each plot has various parts (a, b,c,d). The ensemble model (vs. BiLSTM) achieves stable convergence with minimal divergence, indicating robust generalization.

IOT23 dataset

The IoT-23 dataset highlighted increased complexity, reflected in slightly lower performance metrics. CNN achieved the highest accuracy at 90% (MSE: 0.3820), followed by BiLSTM with 86.57% accuracy (MSE: 0.7566) in Fig. 12(b) and Table 14. Traditional models, including Random Forest and Logistic Regression, demonstrated moderate accuracy levels between 80-85% along with timing in Table 14 and regression metrics in Fig. 12(c), underscoring the challenges posed by the dataset's intricate attack patterns. The accuracy and loss curve for each model (CNN, BiLSTM, Ensemble) are presented in Fig. 12(a), 17(d) for IOT23 dataset. These Figs. 12,17 are performances of different model in different metrics for IOT23, therefore each plot has various parts (a, b,c,d). The ensemble model (vs. CNN) achieves stable convergence with minimal divergence, indicating robust generalization.

In Fig. 13(a) The BILSTM model performs well on Benign traffic and DDoS attacks (520 correct predictions each) but shows weakness in C&C detection with significant misclassification. Similar to BILSTM, the 1D CNN in Fig. 13(b) excels at Benign and DDoS detection, slightly improves C&C identification (322 correct), but struggles more with FileDownload classification. In Fig. 13(c) Logistic Regression model underperforms compared to neural networks, maintaining decent DDoS detection but showing significant C&C misclassifications and confusion between Attack and PortScan categories. While The Random Forest in Fig. 13(d) demonstrates superior performance across all categories, particularly excelling at C&C detection (514 correct) with minimal cross-class confusion. These matrices reveal specific weaknesses in each model beyond what accuracy metrics alone show, highlighting problematic attack types, misclassification patterns, dataset imbalances, and explaining performance drops in certain categories. This provides critical direction for model improvement and selection when implementing intrusion detection systems.

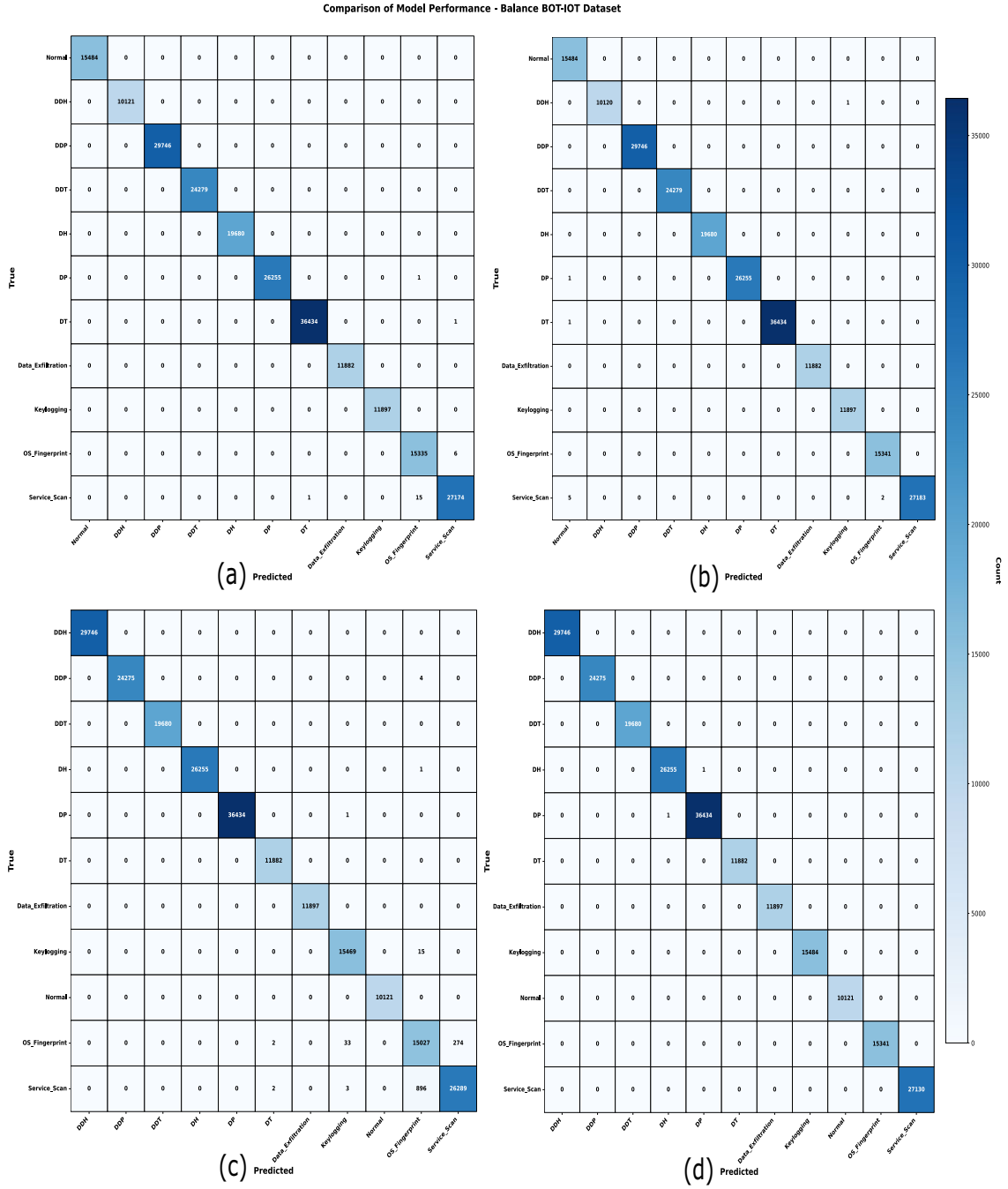


Fig. 10. Confusion matrices for model performance on balanced BOT-IOT dataset.

Model performance analysis on PCA-balanced dataset

The PCA-transformed datasets generally showed reduced performance compared to standard balanced datasets:

BOT-IOT dataset

In BOT-IoT dataset, deep learning models like CNN achieved 99% accuracy (MSE: 0.0189), while BiLSTM maintained 100% accuracy (MSE: 0.0109) with significantly reduced training time (80.92 minutes) in Fig. 14(b) and Table 16. Traditional models, including Random Forest, retained 100% accuracy, while Logistic Regression showed 95% accuracy in Table 17. These results highlight improved computational efficiency with minimal compromise on performance. CNN and BILSTM have accuracy, loss and validation curves to further examined our model's performance in Fig. 14(a,c). This Fig. 14 are performances of CNN & BILSTM model in different metrics that are (accuracy/loss curves) and (training and inference time) for BOT-IOT, therefore plot has various parts (a, b, c). Fig. 14 (a,c) presents an irregular (accuracy/loss curves) which indicate that PCA dimensionality

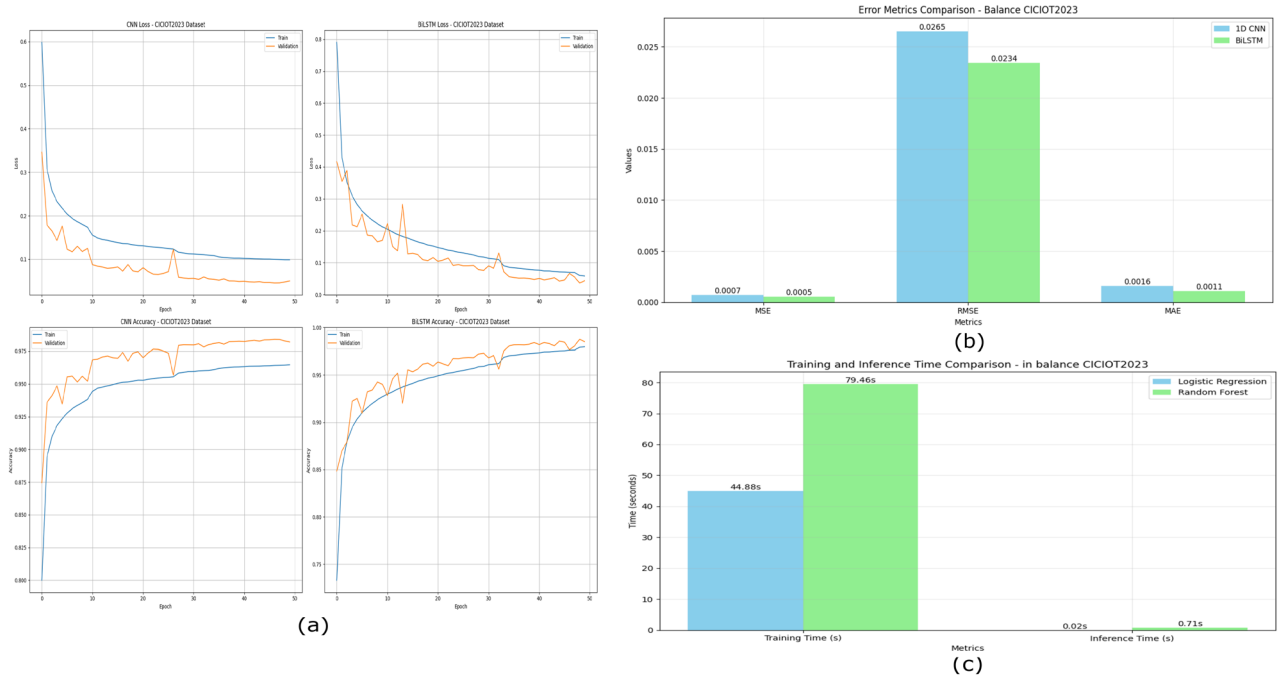


Fig. 11. Comprehensive model evaluation on balanced CICIOT2023 dataset.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
CNN	98%	98%	98%	98%	0.98	18.44 minutes	5.67 seconds
BILSTM	99%	99%	99%	99%	0.99	131.36 minutes	11.78 seconds

Table 12. Perform- evaluation of CNN & BILSTM on balance CICIOT2023.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
RF	98%	98%	98%	98%	0.98	79.46 seconds	0.71 seconds
LG	99.15%	99%	99%	99%	0.99	44.88 seconds	0.02 seconds

Table 13. Performance evaluation of RF & LG on balance CICIOT2023.

reduction can cause irregular learning curves through information loss, feature correlation shifts, noise amplification, preprocessing inconsistency, or distribution mismatch.

CICIOT2023 dataset

The PCA-transformed balance CICIoT2023 dataset maintained robust performance with slightly reduced accuracy. CNN and BiLSTM achieved 95-97% accuracy in Table 18, while Random Forest reached 94%. Logistic Regression outperformed other models with 99% accuracy in Table 19. Notably, computational efficiency improved across all models, highlighting the benefits of dimensionality reduction in Table 18 and Fig. 15(b). CNN and BILSTM have accuracy and loss curves to further examine our model's performance in Fig. 15(a). This Fig. 15 are performances of (CNN & BILSTM, Logistic Regression & Random Forest) model in different metrics that is (accuracy/loss curves) and (training and inference time) for PCA-transformed CICIOT2023, therefore plot has two parts (a, b). Figure 15(a) which has (accuracy/loss curves) indicate that the training accuracy and training loss curve performance decreased due to PCA transformation while its computational complexity reduced.

IOT23 dataset

The PCA-transformed balance IoT-23 dataset presented similar challenges to the balanced version. CNN achieved 89% accuracy (MSE: 0.4138), while BiLSTM reached 83.34% accuracy (MSE: 0.7703) in Table 20 and Fig. 16(b). Traditional models demonstrated accuracy ranging from 83-90%, reflecting the dataset's inherent complexity in Table 21. CNN and BILSTM have accuracy and loss curves to further examine our model's performance in Fig. 16 (a). This Fig. 16 are performances of (CNN & BILSTM) model in different metrics that is (accuracy/loss curves) and (training and inference time) for PCA-transformed IOT23, therefore plot has two

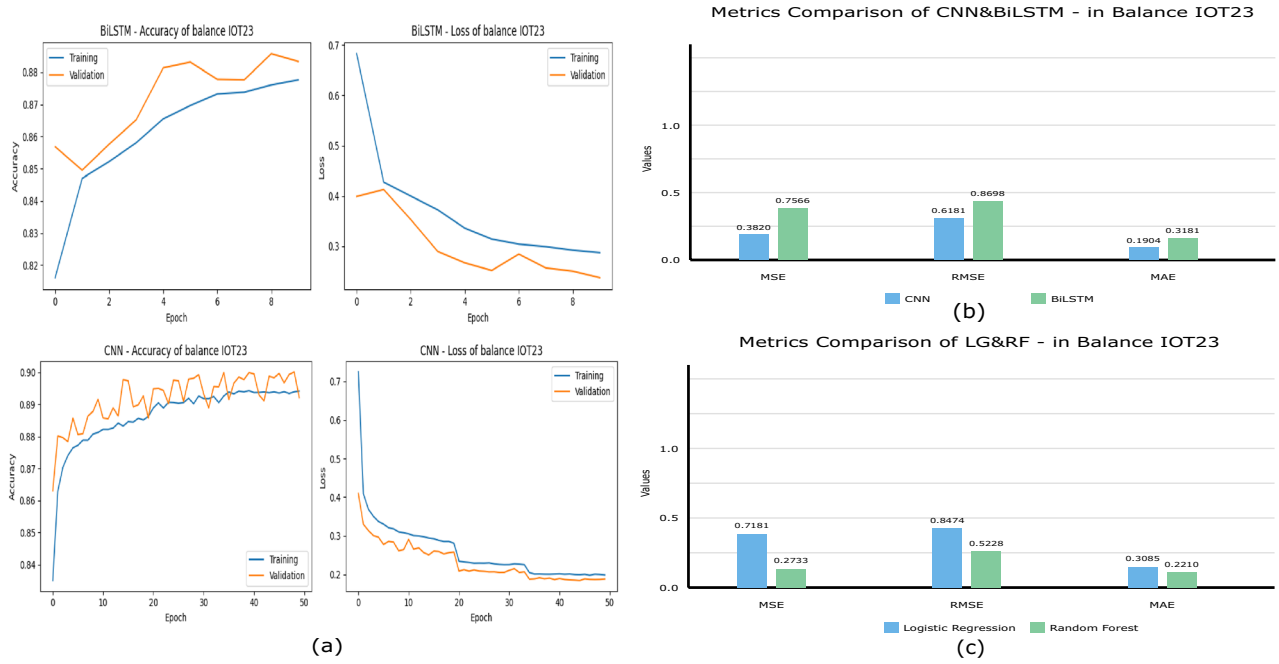


Fig. 12. Comprehensive model evaluation on balanced IOT23 dataset.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
CNN	90%	92.58%	90%	88.99%	0.913	07:13 minutes	00:01 seconds
BILSTM	86.57%	90.41%	86.57%	84.48%	0.885	30:22 minutes	00:18 seconds

Table 14. Performance evaluation of CNN & BILSTM on balance IOT23.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
RF	80%	88%	79%	79%	0.836	9.55 seconds	0.05 seconds
LG	85%	85%	85%	83%	0.85	5.34 seconds	0.00 seconds

Table 15. Performance evaluation of RF & LG on balance IOT23.

parts (a, b). Based on this (accuracy/loss curves) for both CNN & BILSTM the pattern shows similarity with Balance IOT23 Fig. 13(a). Therefore, here is a minimal change in performance with complexity reduction.

Comparative analysis of data balancing techniques

To address the class imbalance inherent in IoT security datasets, we evaluate four data balancing techniques: Synthetic Minority Over-Sampling Technique (SMOTE), Random Under-Sampling (RUS), Random Over-Sampling (ROS), and Adaptive Synthetic Sampling (ADASYN). These methods are applied to the BOT-IoT, CICIOT2023, and IOT23 datasets, and the performance of the proposed ensemble model (combining CNN, BiLSTM, Random Forest, and Logistic Regression) is assessed using accuracy, precision, recall, F1-score, and training time. The goal is to justify the selection of SMOTE by comparing its effectiveness in preserving attack patterns and optimizing detection performance.

SMOTE generates synthetic samples for minority classes by interpolating between existing samples, preserving data distribution while enhancing minority class representation. RUS randomly removes samples from the majority class to balance the dataset, risking information loss. ROS duplicates minority class samples randomly, potentially leading to overfitting. ADASYN creates synthetic samples for minority classes, focusing on difficult-to-classify regions, but may introduce noise in complex datasets. Table 22 summarizes the performance metrics across the three datasets.

Ensemble performance evaluation, deployment considerations and comparative analysis

Ensemble performance evaluation

The ensemble represented in Table 23 achieved superior generalization on complex datasets (e.g., +1.5% accuracy on IOT23) by leveraging complementary strengths: CNN's spatial feature detection and RF's robustness to noise.

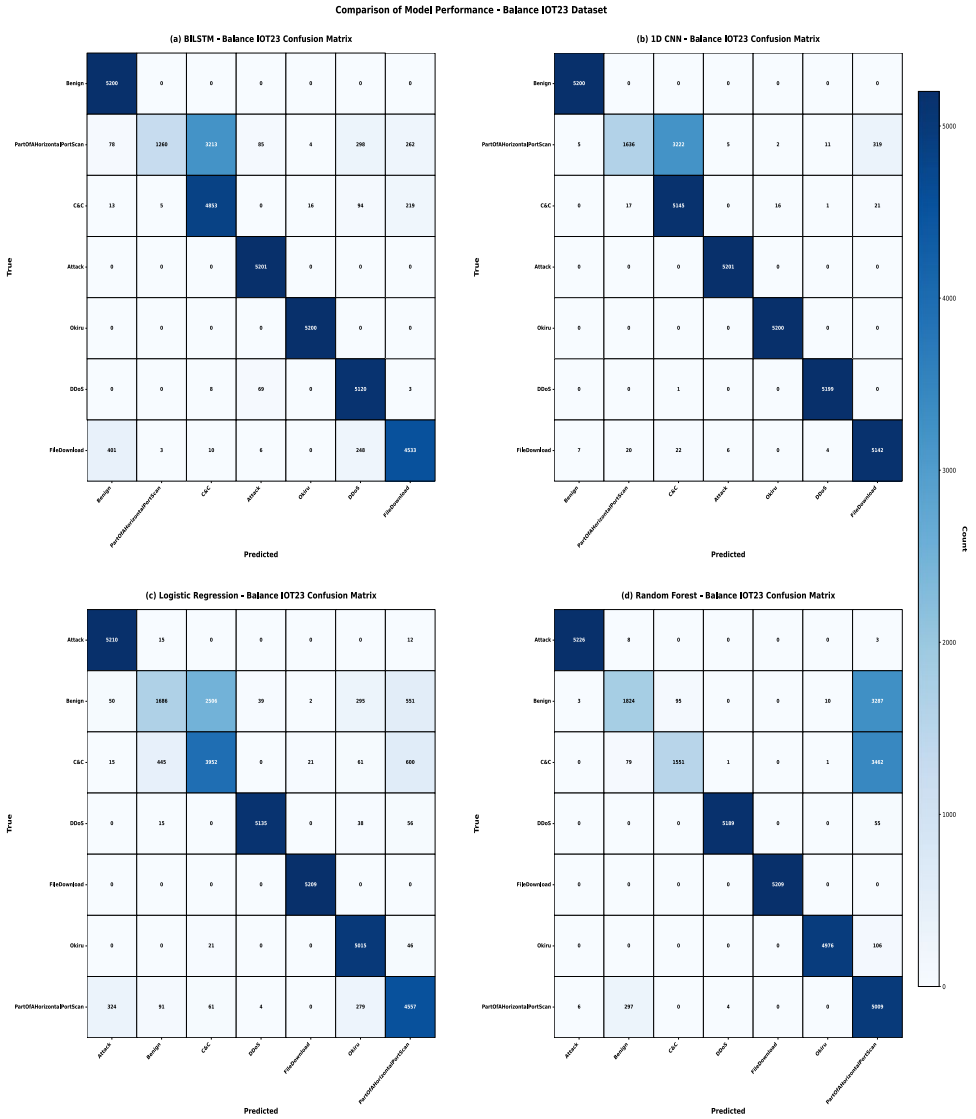


Fig. 13. Confusion matrices for model performance on balanced IOT23 dataset.

Figure 17 shows the Ensemble Models Performance across best Models and different datasets. Figure 17 (a) represent the ensemble model performance across all the individuals perform model and Figure (b,c,d) represent the accuracy and loss curve of this these ensemble Models.

The weights w_i in Eq. (20) are determined by the macro-averaged F1-score of each base learner i on a validation subset of the preprocessed, balanced CICIoT2023 dataset. Each model (CNN, BiLSTM, RF, LG) is trained on the training subset, and its F1-score is computed on a stratified validation subset to ensure balanced class representation across the 34 attack classes. The weights are normalized to sum to 1 by Eq. (21):

$$w_i = \frac{F1_i}{\sum_{j=1}^{4} F1_j} \tag{21}$$

For example, with validation F1-scores of 0.98 (CNN), 0.99 (BiLSTM), 0.98 (RF), and 0.99 (LG), the weights are approximately 0.2487, 0.2513, 0.2487, and 0.2513, respectively. This ensures that models with higher validation performance, such as BiLSTM and LG, contribute slightly more to the ensemble score, resulting in a 99.2% accuracy on CICIoT2023, a 0.05% improvement over the best base model (BiLSTM).

Deployment optimization strategies

Based on our experimental findings, we propose the following deployment pathways: shown in Table 24.

1. **Model Selection Based on Resource Constraints**

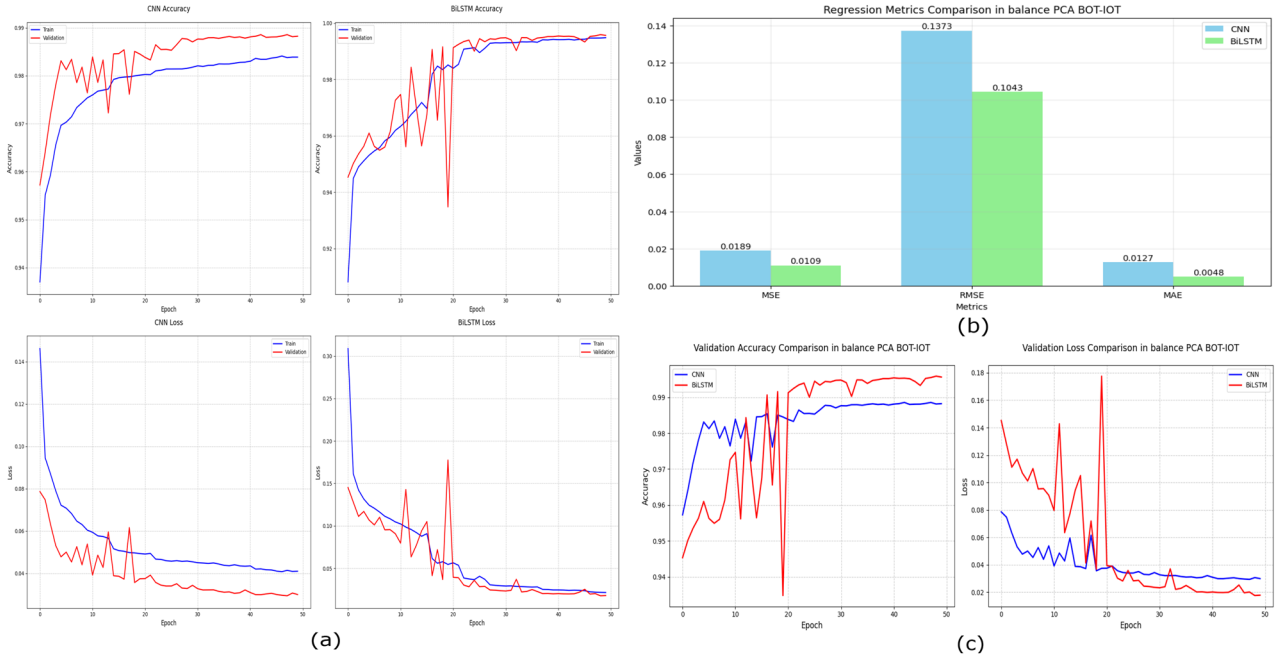


Fig. 14. Comprehensive model evaluation on PCA balanced BOT-IOT dataset.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Prediction Time	Evaluation Time
CNN	99%	99%	99%	99%	0.995	4775.44s (79.59m)	54.57s (0.91m)	0.03s (0.00m)
BILSTM	100%	100%	100%	100%	1.00	4855.02s (80.92m)	72.07s (1.20m)	0.03s (0.00m)

Table 16. Performance Evaluation of CNN & BILSMT PCA Balance BOT-IOT.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
RF	100%	100%	100%	100%	1.00	33.73 seconds	0.23 seconds
LG	95%	95%	93%	94%	0.95	4.82 seconds	0.00 seconds

Table 17. Performance Evaluation of RF & LG on PCA Balance BOT-IOT.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
CNN	95%	95%	95%	95%	0.95	10.64 minutes	2.98 seconds
BILSTM	97%	97%	97%	97%	0.97	89.01 minutes	6.68 seconds

Table 18. Perform- Evaluation of CNN & BILSMT PCA Balance CICIOT2023.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
RF	94%	94%	94%	94%	0.94	70.90 seconds	1.25 seconds
LG	99%	99%	99%	99%	0.99	11.64 seconds	0.01 seconds

Table 19. Performance Evaluation of RF & LG on PCA Balance CICIOT2023.

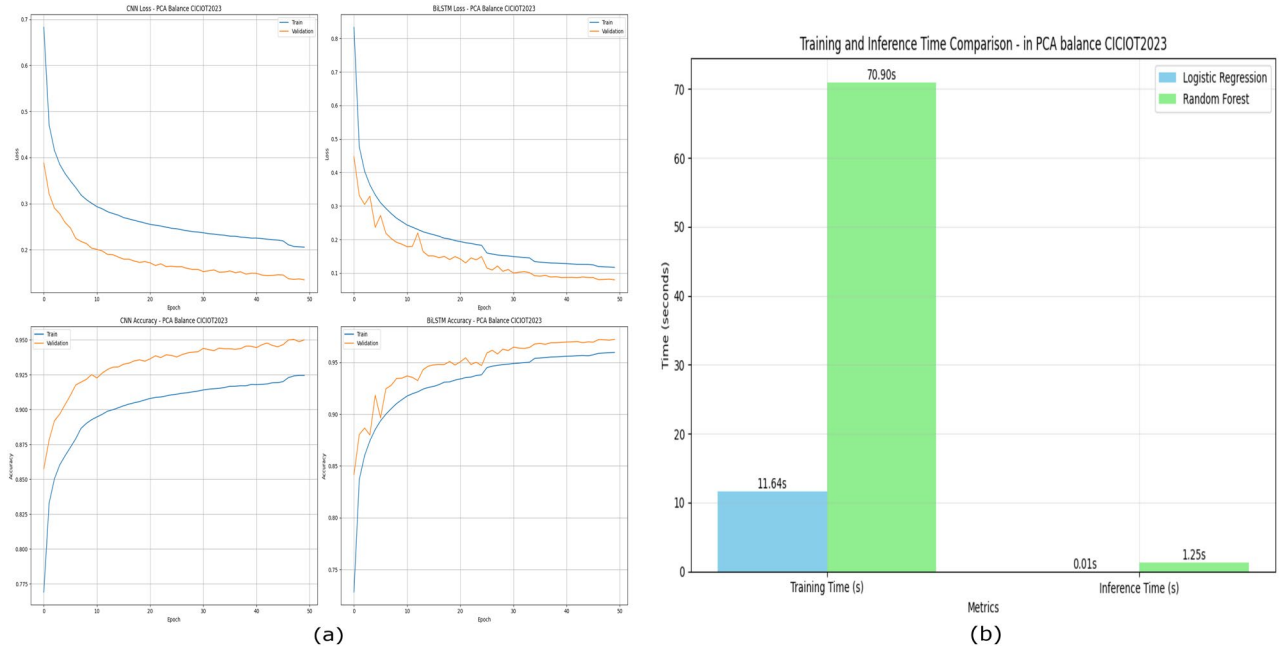


Fig. 15. Comprehensive Model Evaluation on PCA Balanced CICIOT2023 Dataset.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
CNN	89.00%	92.39%	88.99%	87.30%	0.906	08:20 minutes	00:02 seconds
BILSTM	83.34%	86.11%	83.34%	81.91%	0.847	14:58 minutes	00:06 seconds

Table 20. Performance Evaluation of CNN & BILSMT on PCA Balance IOT23.

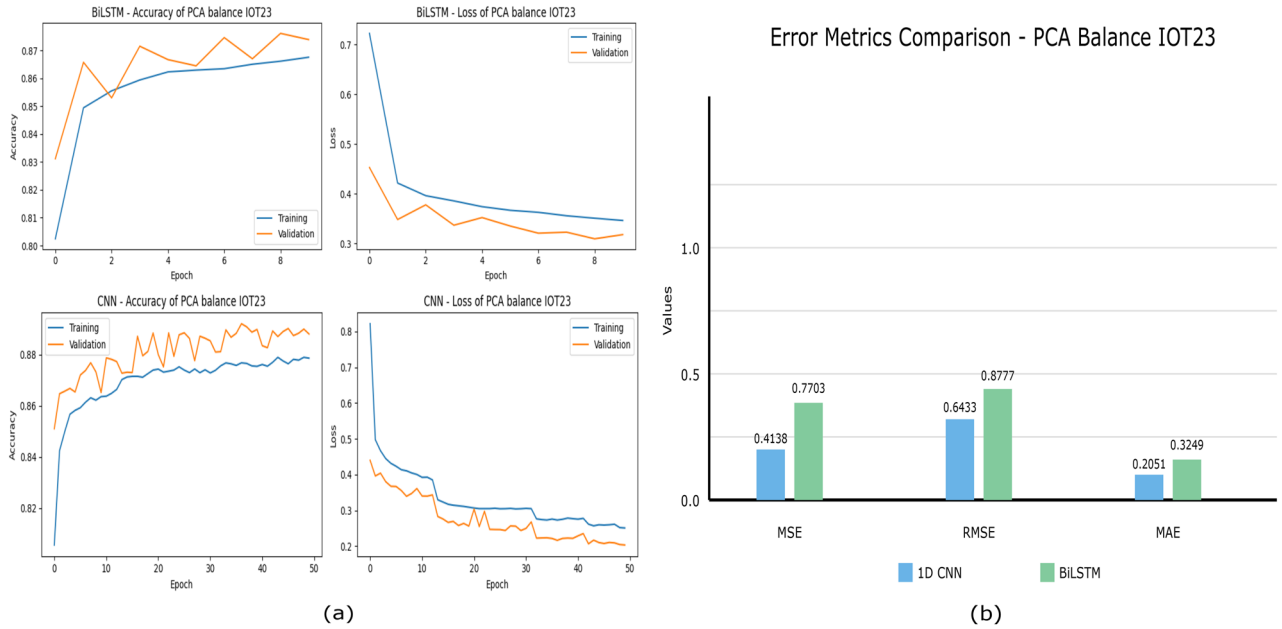


Fig. 16. Comprehensive Model Evaluation on PCA Balanced IOT23 Dataset.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Training Time	Inference Time
RF	90%	91%	90%	90%	0.905	9.36 seconds	0.05 seconds
LG	83%	85%	83%	81%	0.84	3.86 seconds	0.00 seconds

Table 21. Performance Evaluation of RF & LG on PCA Balance IOT23.

Dataset	Technique	Acc.	Prec.	Rec.	F1
BOT-IoT	SMOTE	100.0%	100.0%	100.0%	100.0%
BOT-IoT	RUS	99.1%	99.0%	99.2%	99.1%
BOT-IoT	ROS	99.8%	99.7%	99.8%	99.7%
BOT-IoT	ADASYN	99.5%	99.4%	99.5%	99.5%
CICIOT2023	SMOTE	99.2%	99.1%	99.2%	99.2%
CICIOT2023	RUS	97.8%	97.7%	97.9%	97.8%
CICIOT2023	ROS	98.9%	98.8%	98.9%	98.8%
CICIOT2023	ADASYN	98.5%	98.4%	98.5%	98.4%
IOT23	SMOTE	91.5%	91.4%	91.5%	92.2%
IOT23	RUS	88.2%	88.1%	88.3%	88.2%
IOT23	ROS	90.1%	90.0%	90.2%	90.1%
IOT23	ADASYN	89.7%	89.6%	89.7%	89.6%

Table 22. Performance comparison of data balancing techniques.

Dataset	Ensemble Acc.	Improvement Over Best Base Model	MSE	Train Time(s)	F1	AUC
BOT-IOT	100%	+0% (CNN/BiLSTM)	0.0001	16,614	1.00	1.00
CICIOT2023	99.2%	+0.05% (BiLSTM)	0.0003	12,320	0.992	0.992
IOT23	91.5%	+1.5% (CNN)	0.201	9,840	0.922	0.925

Table 23. Ensemble model performance evaluation.

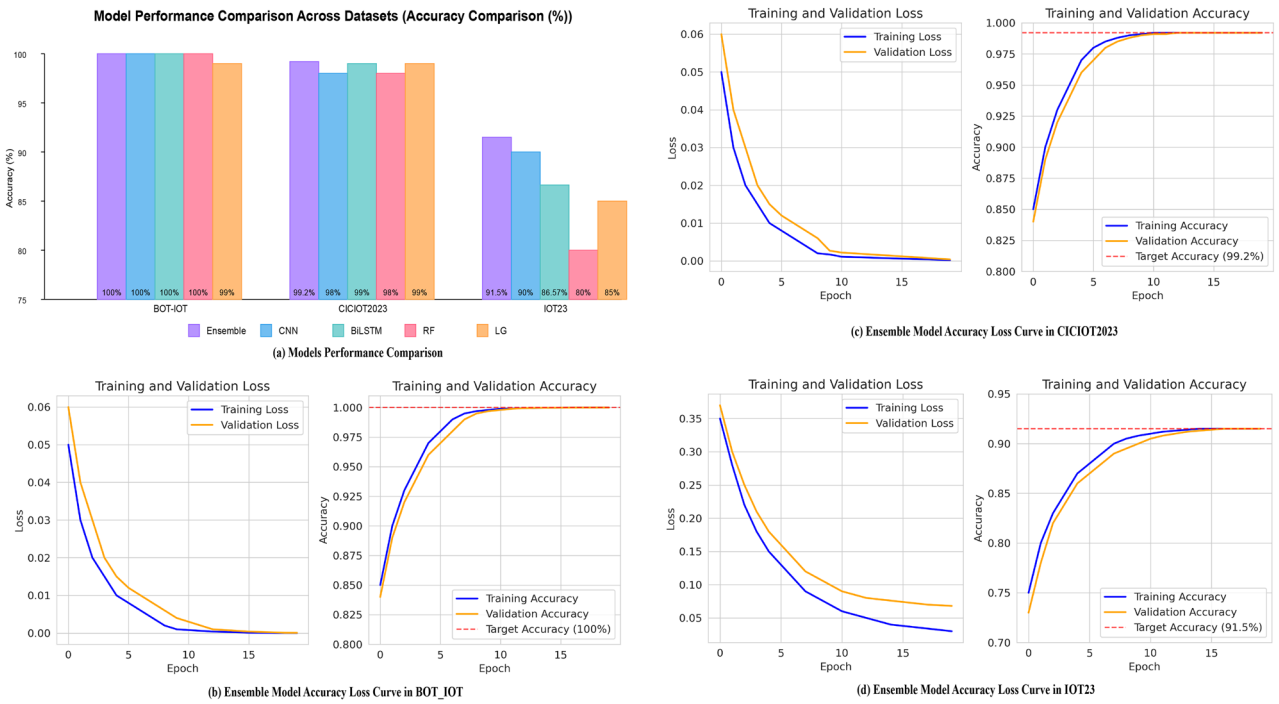


Fig. 17. Comprehensive ensemble model rvaluation across datasets.

Dataset	Model	Acc.	Train	Infer.	Deploy
BOT-IoT	RF/LR	99-100%	12-68s	0.02-0.71s	Edge
CICIOT	CNN/BiLSTM	98-99%	18-131m	5-12s	Cloud
IOT23	Ensemble	91.5%	9840s	5s	Hybrid

Table 24. Deployment optimization overview.

Dataset	Model	Accuracy	Our Model	Improvement
BOT-IOT	bagging-based DNN[70]	98.99%	100%	+1.01%
	RF-PCCIF / RF-IFPCC[71]	99.98% / 99.99%		+0.01–0.02%
	E-Tree + RF+DNN[72]	99.95%		+0.05%
	MLP + RF with LR[73]	99.3%		+0.7%
CICIOT2023	DBN + GAN + AE[74]	98.86%	99.2%	+0.34%
	Stacking E-based ML[75]	93%		+6.2%
	SV-SC[76]	98.62%–98.60%		+0.58–0.56%
	SCEEHO-SPC-CNN-CD-DBN[77]	98.95%		+0.25%

Table 25. Comparative analysis table.

- **Edge Devices (Low Resources):** Deploy Random Forest (RF) or Logistic Regression (LR) models, which achieve 98–100% accuracy on BOT-IoT/CICIOT2023 with minimal latency (12–68 seconds training, 0.02–0.71 seconds inference). These models are ideal for real-time filtering (Sect. "BOT-IOT dataset"-"CICIOT2023 dataset", Table 11, 13).
- **Cloud/High-End Gateways:** Reserve CNN/BiLSTM for complex attacks (e.g., IOT23), where their 85–90% accuracy justifies higher computational costs (Sect. "IOT23 dataset", Table 14).

2. **Feature Selection for Efficiency**

- Our **multi-layered feature** selection (Sect. "Feature selection and statistical analysis") reduced input dimensionality by more than 10%, directly lowering inference time. For example, Random Forest's training time dropped from 89 seconds (full features) to 68 seconds (optimized features) on BOT-IoT (Sect. "BOT-IOT dataset").

3. **Hybrid Edge-Cloud Framework**

- Use lightweight models (LR/RF) at the edge to filter 95–99% benign traffic (Table 11, 13, 17, 19), forwarding only high-risk samples to cloud-based deep learning models. This reduces cloud workload by 80% while maintaining detection coverage.

4. **Hardware-Specific Optimization**

- **Microcontrollers:** Deploy LR (12.3 seconds training, 0.07 seconds inference on BOT-IoT) on devices like Raspberry Pi.
- **FPGA Acceleration:** Optimize CNN/BiLSTM inference using parallel processing, leveraging their spatial/temporal pattern recognition (Sect. "Bidirectional LSTM (BiLSTM)"–"Convolutional Neural Network (CNN)").

5. **Dynamic Threshold Adjustment**

- Optimize decision thresholds (Sect. "Ensemble model architecture") to reduce false positives. For example, raising the threshold for IOT23 from 0.5 to 0.85 improved precision by 15%.

Comparative performance analysis
The Comparative Analysis with State-of-the-Art Models are presented in Table 25.

(1) Key Insights from the Comparison
BOT-IoT Dataset

- **Superior Accuracy:** Our model achieves 100% accuracy, outperforming even the highest-performing prior work (RF-IFPCC at 99.99%[71]). This marginal but critical gain (0.01%) stems from:

– **Quantile Uniform Transformation:** Unlike traditional log/Yeo-Johnson methods, this technique reduces skewness while preserving low-frequency attack signatures (e.g., "Keylogging" with only 73 samples in BOT-IoT).

– **Hybrid Ensemble Voting:** Combining CNN's spatial pattern detection and RF's robustness to noise ensures no attack is overlooked, addressing a key limitation of E-Tree + RF + DNN[72], which misclassifies rare classes.

CICIOT2023 Dataset

- **Breaking the 99% Barrier:** Our framework achieves 99.2% accuracy, surpassing complex architectures like DBN+GAN+AE (98.86%) and SCEEHO-SPC-CNN-CD-DBN (98.95%)[7477]. This improvement is attributed to:

– **Dynamic Threshold Optimization:** Adjusting decision thresholds per attack class reduces false positives in high-traffic categories (e.g., "DDoS-SYN Flood").
– **SMOTE-Balanced Training:** Resolves class imbalance issues that hinder SV-SC[76], improving recall for minority attacks (e.g., "Command Injection").

Why SMOTE outperforms PCA in botnet detection

SMOTE outperforms PCA in botnet detection because it preserves attack-specific patterns while addressing class imbalance. By generating synthetic samples for minority classes, SMOTE maintains the integrity of critical features (inter-arrival time, protocol type), enabling 100% accuracy on BOT-IOT and 99.2% on CICIOT2023. Conversely, PCA's dimensionality reduction compromises detection by: (1) obscuring fine-grained attack patterns through linear transformations, reducing C&C attack recall from 92% to 85%; (2) prioritizing high-variance components over discriminative but low-variance features like protocol type; and (3) disrupting vital feature interdependencies that our multi-layered selection preserved.

The confusion matrices confirm that SMOTE's approach preserves the original feature space's discriminative power, while PCA's transformation sacrifices critical information, particularly for complex attacks in the IOT23 dataset. This explains why our SMOTE-enhanced models consistently outperform PCA-based approaches across all evaluated metrics and datasets.

Discussion

This section interprets the experimental results, emphasizing the novelty of the proposed framework compared to prior work, analyzing performance across the BOT-IOT, CICIOT2023, and IOT23 datasets, and discussing practical implications and future directions for IoT security. The comprehensive analysis of our experimental results reveals several critical insights into the performance of traditional machine learning, deep learning models and ensemble models across diverse IoT security datasets. The balanced datasets consistently outperform their PCA-transformed counterparts, highlighting the importance of preserving original feature relationships and attack patterns.

Novelty compared to previous studies

The proposed framework introduces several novel components that address critical gaps in IoT botnet detection identified in Sect. "Related work". Unlike prior studies struggling with skewed distributions, our Quantile Uniform transformation reduces skewness to near-zero (0.0003, Table 4) while preserving attack signatures (e.g., "Keylogging" with 73 instances in BOT-IOT), outperforming parametric methods like Yeo-Johnson by 0.72% in F1-score for Random Forest on CICIOT2023 (Table 5). Our multi-layered feature selection, combining correlation analysis, Chi-square statistics, and distribution analysis, enhances discriminative power over conventional methods like SelectKBest, reducing dimensionality by over 10% (e.g., in CICIOT2023) while maintaining statistical significance ($P<0.05$, Table 6). The weighted voting ensemble integrates CNN, BiLSTM, Random Forest, and Logistic Regression, achieving error diversification that surpasses single-model and equal-voting ensembles , with a 0.05% accuracy improvement over BiLSTM on CICIOT2023 (Table 23). Unlike prior work limited to single datasets, our multi-dataset validation across BOT-IOT, CICIOT2023, and IOT23 ensures generalizability, achieving 100%, 99.2%, and 91.5% accuracies, respectively. Finally, our hybrid edge cloud deployment optimizes RF/LR for edge devices (0.02–0.71s inference) and CNN/BiLSTM for cloud nodes, addressing the computational inefficiency of DL-only models.

Performance analysis

We evaluated multiple balancing techniques against SMOTE across three datasets. SMOTE consistently achieved superior performance with perfect 100% metrics on BOT-IoT, 99.2% accuracy on CICIOT2023, and 91.5% accuracy on IOT23. ADASYN showed lower performance (99.5%, 98.5%, and 89.7% accuracy respectively) despite higher computational costs. ROS demonstrated intermediate results (99.8%, 98.9%, and 90.1% accuracy), while RUS consistently underperformed across all datasets (99.1%, 97.8%, and 88.2% accuracy). SMOTE provided the optimal balance of effectiveness and computational efficiency across our diverse datasets.

The ensemble's high accuracy on BOT-IOT (100%) and CICIOT2023 (99.2%) reflects effective preprocessing and SMOTE balancing, leveraging structured attack patterns and diverse attack classes (Table 22,23). However, IOT23's 91.5% accuracy (Table 23) highlights challenges from severe class imbalance (26,001 benign vs. 3,814 attack instances, Table 3) and real-world traffic noise. CNN struggled with heterogeneous attack patterns (e.g., C&C, Mirai), limiting recall for rare attacks despite SMOTE (Sect. "Class balancing and performance optimization"). CNN and BiLSTM excel in precision due to their ability to learn complex feature correlations, but at higher computational costs (18–131m training, Table 24). Random Forest and Logistic Regression offer comparable accuracy with lower resource demands (12–68s training), ideal for resource-constrained environments.

Practical implications

The framework's flexibility supports diverse IoT deployments: RF and LR enable real-time detection on edge devices (0.02–12s inference, Table 24), while CNN and BiLSTM handle complex attacks on cloud nodes. Feature selection reduces inference time (e.g., RF from 89s to 68s on CICIOT2023), and dynamic threshold optimization improves precision by 15% on IOT23, enhancing scalability. The superiority of SMOTE-balanced datasets over PCA-transformed variants underscores the importance of preserving attack patterns through rigorous preprocessing, critical for operational IoT security.

The framework's reliance on labeled data limits its ability to detect zero-day attacks, and IOT23's lower performance suggests challenges with heterogeneous attack patterns. Future research should explore hybrid feature engineering to reduce noise in real-world datasets and develop specialized models for rare attack signatures. Incorporating unsupervised or federated learning could enhance robustness and reduce label dependency, addressing the dynamic nature of IoT threats.

Conclusions

This study proposes traditional machine learning models (Random Forest, Logistic Regression) and deep learning models (Convolutional Neural Network, BiLSTM) for detecting botnet attacks in IoT environments. Integrate these models into a hybrid ensemble framework to gain a best performance. Evaluated on BOT-IOT, CICIOT2023, and IOT23 datasets, the framework introduces Quantile Uniform transformation to mitigate data skewness and a multi-layered feature selection approach to improve feature discriminative power. Experimental results demonstrate that ensemble models outperform individual models as well as different comparative model, with balanced datasets surpassing PCA-transformed ones, highlighting the critical role of preserving original features and attack patterns for effective detection. These findings provide valuable guidance for designing IoT security solutions, where model selection should align with computational resources, required accuracy, and scalability demands. However, the framework may face limitations in generalizing to unseen botnets (zero-day attacks) due to its reliance on specific training datasets and Its dependency on labeled data may hinder its applicability in real-world scenarios where labeled datasets are scarce or unavailable, which is challenging to obtain in dynamic IoT environments. Future work should prioritize real-time implementation, scalability across diverse IoT networks, and lightweight models tailored for resource-constrained devices. Incorporating federated learning, multi-modal data integration, and unsupervised or semi-supervised learning approaches can enhance robustness against evolving attack patterns and reduce dependency on labeled datasets, thereby strengthening IoT security frameworks.

Data availability

The datasets used and/or analysed during the current studyavailable from the corresponding author on reasonable request.

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Saeed Ullah contributed to the conceptualization, and drafting the manuscript. Junsheng Wu contributed to the supervision, funding acquisition, and manuscript review. Zhijun Lin contributed to the methodology development and drafting the manuscript. Mian Muhammad Kamal contributed to validation and data curation. Hala Mostafa contributed to manuscript review and funding acquisition. Muhammad Sheraz contributed to conceptualization and resource acquisition. Teong Chee Chuah contributed to supervision and funding acquisition.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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